How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning

by

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Abstract

Prompts for in-context learning (ICL) often describe a task as well as demonstrate it through examples. Yet how descriptions work, especially in relation to demonstrations, remains unclear. Through the Bayesian view of ICL, we characterize descriptions as informative priors by their specificity and reliability, and borrow the effective sample size (ESS) from Bayesian statistics to measure their worth in demonstrations. Our theoretical results on in-context linear regression tasks show that a reliable description's ESS grows with its specificity and with the task's dimension. However, the ESS drastically falls if the description may be irrelevant: a one-in-ten chance of irrelevance cuts a precise description's ESS from 117 to 13.5. We then translate the setting into experiments on Transformer networks and pretrained language models, including frontier ones. Meta-trained Transformers come close to the Bayes-optimal ESS and adapt to unreliable descriptions when trained on them. Pretrained language models, however, surprisingly show little regard for descriptions when answering without reasoning, and closely match the Bayesian reference only when reasoning is enabled. This effect is mostly robust to different levels of model capability.

Preface

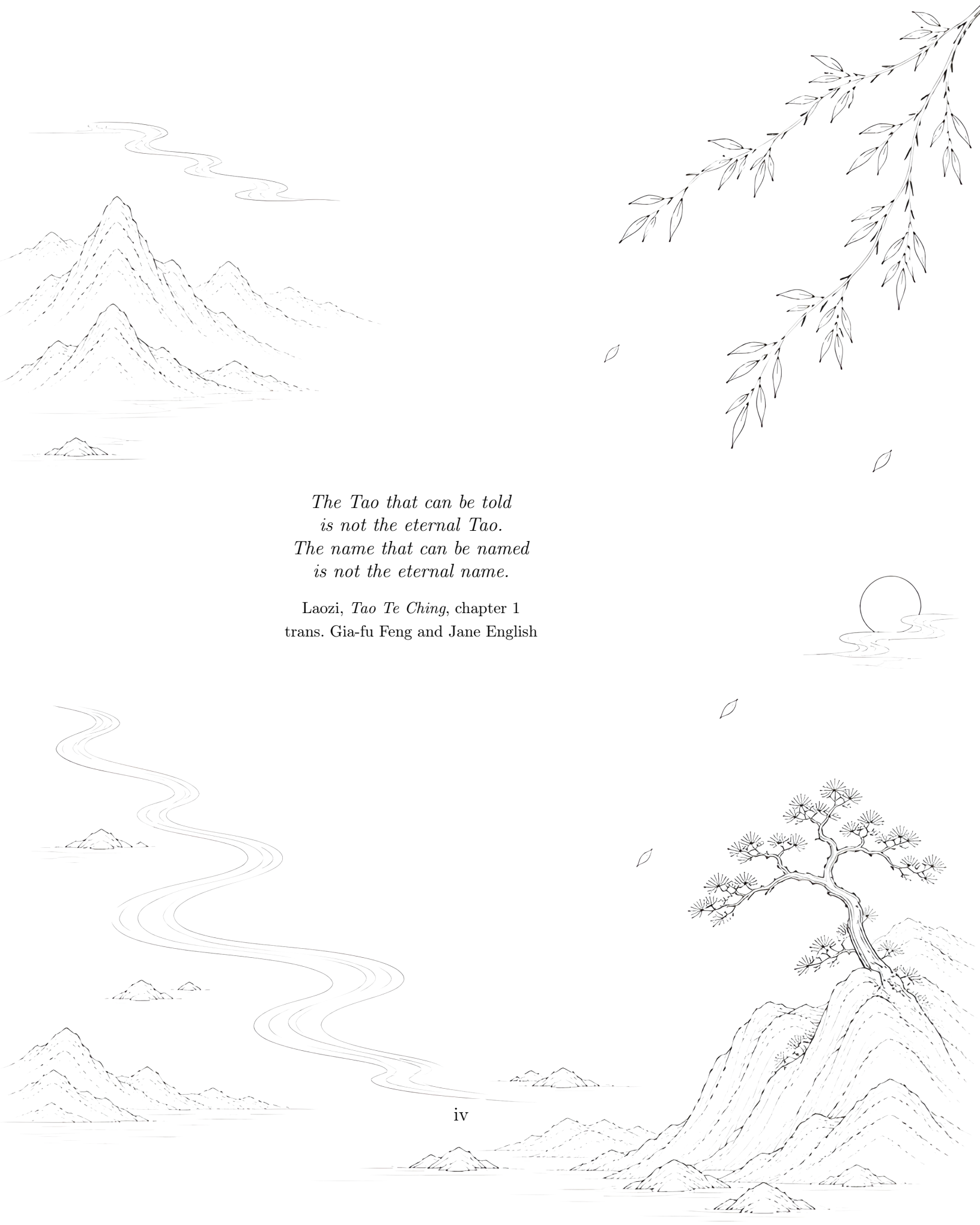
This thesis is an original work by Trung Quan Nguyen, carried out under the supervision of Professor Linglong Kong and Professor Bei Jiang. **[TODO: Describe any contribution of the supervisors or others to the research questions, design, analysis, or writing, or state that there was none.]**

The thesis is based on the manuscript "How Many Demonstrations Is a Task Description Worth? Specificity and Reliability in In-Context Learning" by Bruce Quan Nguyen, the name under which the author writes, prepared in the format of the International Conference on Machine Learning. Chapters 1 and 3 to 8 and Appendix A follow that manuscript's text, with its appendices placed in the chapters they support and some added detail; Section 8.2 is new. Chapter 2 is new, except that Sections 2.2, 2.5, 2.8, and 2.9 take the manuscript's review of related work. **[TODO: State whether the manuscript has been submitted or published; if so, give the full citation.]**

The research involved no human participants, animals, or personal data, so no research ethics approval was required.

Large language models are the object of study in Chapter 7. Separately, generative AI assistants were used in carrying out this research. **[TODO: Confirm and adjust this draft: Anthropic's Claude Code and OpenAI's Codex were used to write and run experiment code, to check derivations, and to draft and edit text; the author designed the study and reviewed, and takes responsibility for, all code, results, and text. Confirm that the program and supervisory committee permitted this use.]**

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The Tao that can be told
is not the eternal Tao.
The name that can be named
is not the eternal name.

Laozi, *Tao Te Ching*, chapter 1
trans. Gia-fu Feng and Jane English

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Chapter 1

Introduction

Large language models are capable of in-context learning (ICL): given a prompt that specifies a task through a natural-language description, demonstrations, or both, they can predict the output for a new query input (Brown et al., 2020). In practice, prompts consist mostly of system instructions, user instructions, skill files, and earlier conversation turns, which typically describe the task rather than demonstrate it (Mao et al., 2025; Patsakis et al., 2026). The balance shifts further as LLMs carry out increasingly long-horizon tasks (Kwa et al., 2025), for which demonstrations are costly to provide.

Theories of ICL, however, have mostly focused on prompts made of demonstrations alone (Xie et al., 2022; Garg et al., 2022; Akyürek et al., 2023). The few that include descriptions study only specific types, such as a descriptor that describes the input distribution (Huang & Ge, 2025), a set of candidate hypotheses (Lin et al., 2025), or related datasets supplied as context (Zhu et al., 2026); none offers a systematic treatment of descriptions in general. We therefore ask: how can we characterize natural-language descriptions to formally understand and analyze them? And how do descriptions compare, combine, and interact with demonstrations?

To address these questions, we study d-dimensional Gaussian linear regression, a standard controlled setting for ICL (Garg et al., 2022; Panwar et al., 2024). A description supplies information about the regression coefficients before demonstrations are observed. A mixture model separates two kinds of uncertainty: conditional on relevance, the description specifies a Gaussian distribution with precision c; with probability $1 - p$, it is independent of the task. We thus characterize a description by two quantities: its *specificity*, set by c, and its *reliability*, the prior probability p that

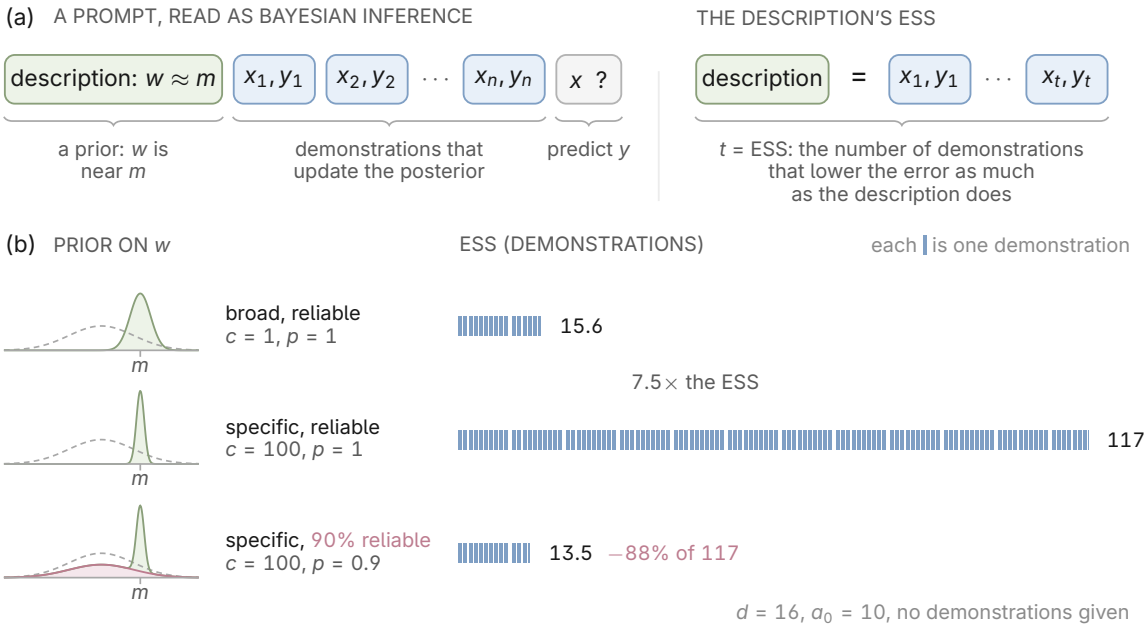


Figure 1.1: **Measuring a task description in demonstrations.** (a) A description provides information about a latent task w, and demonstrations supply additional evidence. Its predictive effective sample size (ESS) is the number of additional demonstrations a learner without the description needs to match its prediction error with the description. The schematic on the right illustrates $n = 0$. (b) Description-conditioned task distributions and their Bayes-optimal ESS at $n = 0$, $d = 16$, and $a_0 = 10$. Specificity c is the precision of the relevant component; reliability p is its prior probability. Pink denotes the irrelevant component. Increasing specificity greatly increases ESS when $p = 1$, whereas a 10% probability of irrelevance sharply limits this benefit.

it is relevant (Figure 1.1).

To compare descriptions with demonstrations, we adapt sample-size measures of prior impact and predictive performance (Wiesenfarth & Calderazzo, 2020; Reimherr et al., 2021; Le Scao & Rush, 2021; Selby et al., 2025). For a given learner and prediction loss, a description's *predictive effective sample size* (ESS) at n demonstrations is the additional number of demonstrations needed without the description to reach the same expected error. Computing ESS for both the Bayes-optimal predictor and trained models distinguishes the information available under the statistical model from the benefit realized by a learner.

We test the analysis with Transformers trained from scratch on the Gaussian model and with pretrained language models prompted on numerical mean-estimation and semantic classification tasks. The Transformers learn a description token's precision and reliability from the training distribution. The language models receive probabilistic descriptions and policy constraints in text.

1.1 Main Findings

Our main findings are as follows:

- **Precision and dimension determine the value of reliable descriptions.** With fixed positive integer precision c, zero-demonstration ESS approaches $c + d + 1$ as the baseline prior becomes diffuse. A partial description can become more valuable as demonstrations constrain the directions it leaves unspecified (Chapter 4).
- **Uncertain relevance limits zero-demonstration ESS.** Any fixed positive probability of irrelevance imposes a finite ceiling as precision grows. For $d = 16$, baseline variance $a_0 = 10$, and $c = 100$, reducing reliability from 1 to 0.9 reduces ESS from approximately 117 to 13.5. Demonstrations can provide evidence about relevance (Chapter 5).
- **Description use depends on training and prompt specifications.** Meta-trained Transformers approach the Bayes-optimal ESS, while their response to reliability shifts reflects training. For most pretrained language models tested, enabling reasoning improves the use of stated precision. For two models, explicitly specifying the irrelevant-component distribution also produces predictions consistent with Bayesian model averaging in numerical and semantic tasks (Chapters 6–7).

1.2 Outline

The results of Chapters 3 to 7 also appear in the manuscript described in the Preface. This thesis presents them in full, together with the background they rest on, complete proofs, and all experiments. The rest of the thesis is organized as follows. Chapter 2 reviews the background, namely in-context learning and its Bayesian interpretation, the effective sample size of a prior, and robust mixture priors, together with related work. Chapter 3 presents the framework: tasks and demonstrations, descriptions as

uncertain task information, prediction error, and predictive ESS. Chapter 4 derives the value of a reliable description, with its proofs and the analysis of partial descriptions. Chapter 5 derives the cost of unreliability, with its proofs and the analysis of the Bayesian predictor assuming relevance under misspecification. Chapter 6 tests meta-trained Transformers, with their training details and further networks. Chapter 7 tests pretrained language models on numerical prediction tasks with explicit probabilistic descriptions and on a semantic classification task, with the evaluation details and additional results. Chapter 8 discusses the findings and their limitations, and outlines future work. Appendix A reports the language-model tables at every evaluated demonstration count.

Chapter 2

Background

In this chapter, we review the ideas that the rest of the thesis builds on. Sections 2.1 to 2.3 introduce in-context learning, its interpretation as Bayesian inference, and in-context linear regression, a setting in which that interpretation can be computed exactly. Section 2.4 defines the squared error by which we compare learners and gives the error of the Bayesian predictor. Sections 2.5 and 2.6 present two tools from Bayesian statistics that we later apply to descriptions: the effective sample size of a prior and the robust mixture prior. Finally, Sections 2.7 to 2.9 review the work most closely related to ours.

2.1 In-Context Learning

A language model takes a sequence of tokens, the prompt, and predicts what comes next. Brown et al. (2020) showed that large language models can perform a task specified in the prompt, without any update to their weights. This ability is called in-context learning (ICL). Brown et al. (2020) distinguished three kinds of prompts. A zero-shot prompt gives only a natural-language description of the task. One-shot and few-shot prompts add one or several *demonstrations*, solved examples of the task, before the query that the model must answer. We follow their terminology. A demonstration is one input together with its label, and apart from the query we call everything else in the prompt a *description*, whether it is written as a statement about the task or as a command. A prompt can contain both, and our question is how much the description adds to the demonstrations.

How models use the demonstrations is itself an active question. Pretrained models can perform well even when the labels of the demonstrations are replaced by random ones (Min et al., 2022), which suggests that they use demonstrations partly to recognize a task they already know. Larger models, however, follow flipped in-context labels more readily (Wei et al., 2023). Lin & Lee (2024) explain both behaviors, retrieving a known task and learning a new one, within a single Bayesian model. A description raises the same question from the other side. Does a model treat it as information about the task, to be weighed against the demonstrations, or does it use it in some other way?

2.2 In-Context Learning as Bayesian Inference

Theoretical work studies ICL as Bayesian inference over latent tasks (Xie et al., 2022; Panwar et al., 2024), as a learning algorithm implemented in a forward pass (Akyürek et al., 2023; von Oswald et al., 2023), and as meta-learning across tasks (Garg et al., 2022; Müller et al., 2022). Meta-trained predictors can approximate Bayesian inference under suitable training conditions (Genewein et al., 2025), and the learning algorithms that Transformers implement in context approach the Bayes-optimal estimator as the networks grow (Akyürek et al., 2023). Agreement can change with training-task diversity or new function classes (Raventós et al., 2023; Panwar et al., 2024), and language-model ICL can violate Bayesian consistency (Falck et al., 2024). We adopt the Bayesian view, because it extends naturally from demonstrations to descriptions (Chapter 3).

Xie et al. (2022) proposed that ICL can be understood as implicit Bayesian inference. Suppose that pretraining text is generated from latent concepts. A model that predicts such text well must then infer the concept behind each document, and a prompt of demonstrations is evidence about the concept behind the prompt. In this view, a task is a latent parameter w, and a prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query x. The ideal learner predicts the query's label with the posterior predictive distribution

$$f(y \mid x, D_n) = \int f(y \mid x, w)\, f(w \mid D_n)\, dw. \tag{2.1}$$

This view becomes testable when the prior over tasks is known. Consider a Transformer trained on prompts whose tasks are sampled from a known prior. Training

on such prompts is called meta-training: the network is trained on many tasks and must infer from the context which task a prompt comes from. The posterior predictive minimizes the expected log loss over the prior (Section 2.4), and this expected log loss is exactly the training objective. A well-trained network therefore approximates the posterior predictive, whose mean is the Bayes-optimal prediction under squared error (Section 2.4) (Müller et al., 2022). For regression with a Gaussian task prior, meta-trained Transformers behave like the Bayesian predictor, which is ridge regression (Akyürek et al., 2023; Raventós et al., 2023; Panwar et al., 2024), provided that they are trained on enough distinct tasks (Raventós et al., 2023). The Bayesian predictor is then a benchmark against which a network can be measured, and Chapter 6 uses it in this way.

2.3 In-Context Linear Regression

Garg et al. (2022) introduced in-context linear regression as a controlled test of ICL. Each prompt has its own task vector $w \sim \mathcal{N}(0, I_d)$, its inputs are drawn as $x_i \sim \mathcal{N}(0, I_d)$, and its labels are $y_i = w^\top x_i$. Garg et al. (2022) found that a Transformer trained from scratch on such prompts predicts the query's label about as well as least squares. Later work added label noise, $y_i = w^\top x_i + \epsilon_i$ (Akyürek et al., 2023; Raventós et al., 2023). Raventós et al. (2023) also varied the number of distinct tasks seen in training. With few tasks, the network behaves like the Bayesian predictor for a prior over those tasks alone. With enough tasks, it behaves like ridge regression, the Bayesian predictor for the Gaussian prior.

With Gaussian noise and a Gaussian prior, Bayes-optimal prediction has a closed form (Gelman et al., 2013, Section 14.8). We measure labels in units of their noise, so that $y_i \sim \mathcal{N}(w^\top x_i, 1)$. Let the prior be $w \sim \mathcal{N}(m, \Lambda^{-1})$, with center m and precision Λ. Let $X_n \in \mathbb{R}^{n \times d}$ contain the demonstration inputs as rows, and write $W_n = X_n^\top X_n$. The posterior over w is Gaussian, with covariance and mean

$$\Sigma = \big(\Lambda + W_n\big)^{-1}, \qquad \mu = \Sigma\big(\Lambda m + X_n^\top y_{1:n}\big), \tag{2.2}$$

and the posterior predictive for the query is

$$y \mid x, D_n \sim \mathcal{N}\big(\mu^\top x,\ 1 + x^\top \Sigma x\big). \tag{2.3}$$

The predictive variance has two parts. The first, 1, is the label noise. The second, $x^\top \Sigma x$, is the remaining uncertainty about the task in the direction of the query. Under

the baseline prior $\mathcal{N}(0, a_0 I_d)$, the posterior mean $\mu = (W_n + a_0^{-1} I_d)^{-1} X_n^\top y_{1:n}$ is the ridge regression estimate with penalty a_0^{-1}. Equation (2.2) also shows that a prior acts like additional data. Each demonstration adds $x_i x_i^\top$ to the posterior precision, and the prior adds Λ. Chapter 3 uses this fact to give a description a prior of its own, and in Chapter 4 a reliable description replaces the baseline precision a_0^{-1} with c.

2.4 Squared Error

A predictor sees the prompt and gives a number $\hat{y}$ for the query's label. We score it by squared error. The label is $y = w^\top x + \epsilon$, where the noise ϵ has mean 0 and variance 1 and is independent of the prompt and the task. The expected squared error therefore splits into two parts:

$$\mathbb{E}\big[(\hat{y} - y)^2\big] = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big] + 1. \tag{2.4}$$

The second part is the noise variance, which is the same for every predictor. Following Garg et al. (2022), we therefore score a predictor by the first part alone, its *prediction error*, or *risk*,

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big],$$

the expected squared distance from the noiseless label $w^\top x$. A predictor that knows the task predicts $w^\top x$ and has zero error.

Given the prompt, the task is uncertain. For any prediction $\hat{y}$ computed from the prompt,

$$\mathbb{E}\big[(\hat{y} - w^\top x)^2 \mid x, D_n\big] = \big(\hat{y} - \mathbb{E}[w^\top x \mid x, D_n]\big)^2 + \mathrm{Var}(w^\top x \mid x, D_n).$$

The first term vanishes only when $\hat{y}$ is the posterior mean of $w^\top x$. When tasks are drawn from the prior, the posterior mean therefore has the smallest error of any predictor. We call it the Bayesian predictor and its error Bayes-optimal. It is also the mean of the posterior predictive (2.1). Its error is the expected posterior variance of $w^\top x$. In linear regression, the posterior is Gaussian with covariance Σ (Equation (2.2)), so the posterior variance of $w^\top x$ is $x^\top \Sigma x$, the part of the predictive variance (2.3) that comes from uncertainty about the task. The query input is independent of the demonstrations, with $\mathbb{E}[xx^\top] = I_d$, so the error of the Bayesian predictor is

$$R = \mathbb{E}\big[x^\top \Sigma x\big] = \mathbb{E}\,\mathrm{tr}\,\Sigma, \tag{2.5}$$

the expected trace of the posterior covariance. The error is zero when the posterior has concentrated on the task, and it grows with the uncertainty about the task that remains.

Two standard facts give this trace in the cases that later chapters need. Chapter 3 writes $R^{\mathrm{demo}}(n)$ for the error with n demonstrations and no description. With fewer demonstrations than dimensions, $n < d$, the demonstrations say nothing about $d - n$ directions, so under the baseline prior $\mathcal{N}(0, a_0 I_d)$ we have $R^{\mathrm{demo}}(n) \geq (d-n)\, a_0$. Under a flat prior, the limit $a_0 \to \infty$, and with $n \geq d + 2$, the posterior covariance is W_n^{-1}, the inverse of a Wishart matrix, and the error is $\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = d/(n - d - 1)$ (Muirhead, 1982). Chapter 4 builds its limits for a reliable description (Proposition 4.1) from these two facts.

A predictor that gives a density $g(y \mid x, D_n)$ rather than a number can instead be scored by its log loss, $-\log g(y \mid x, D_n)$. The expected log loss is smallest for the true conditional density of y given the prompt (Cover & Thomas, 2006), which is the posterior predictive when tasks are drawn from the prior. Meta-trained networks are often trained with this loss (Section 2.2), and so are the networks of Chapter 6. The mean of the posterior predictive is the posterior mean, so a network that minimizes log loss and outputs a predictive mean can be scored by squared error. We use squared error throughout for two reasons. First, it needs only the learner's answer, a number, so it applies to a language model through the number that it writes (Chapter 7). Second, the Bayes-optimal error (2.5) has the exact forms above, so the theory in Chapters 4 and 5 can work with exact errors.

Squared error also splits exactly over the components of a mixture. Suppose that a latent indicator Z selects the component that generates the task. By the law of total variance, the posterior variance is the sum of two parts:

$$\begin{aligned} \mathrm{Var}(w^\top x \mid x, D_n) = \mathbb{E}\big[\, \mathrm{Var}(w^\top x \mid x, D_n, Z) \mid x, D_n \big] \\ + \mathrm{Var}\left(\mathbb{E}[w^\top x \mid x, D_n, Z] \mid x, D_n\right). \end{aligned} \tag{2.6}$$

The first part is the average variance within the components. The second is the variance of the prediction across the components. Chapter 5 applies this decomposition to a description that may be irrelevant. There, the first part gives the risk of the Bayes-optimal predictor that also observes Z, and the second is the excess Bayes risk due to unobserved relevance.

2.5 The Effective Sample Size of a Prior

Bayesian statistics measures the information in a prior by its effective sample size (ESS), the number of observations that would carry the same information. In the conjugate normal model, this number is exact. Suppose that the observations are $y_i \sim \mathcal{N}(w, \sigma^2)$ and the prior is $w \sim \mathcal{N}(m, \sigma^2/c)$. After n observations with mean $\bar{y}$, the posterior is

$$w \mid y_{1:n} \sim \mathcal{N}\Big(\frac{c\,m + n\bar{y}}{c+n},\ \frac{\sigma^2}{c+n}\Big).$$

This is the posterior that $c+n$ observations would give under a flat prior if c of them averaged m. The prior is therefore worth c observations, which is its precision measured in units of the precision of one observation. Chapter 7 uses this model for its prompt-based Bayesian reference, ignoring negligible boundary effects. The specificity c of a description, defined in Chapter 3, is the same conjugate-prior ESS, counted in each direction in units of direct observations of the task. Chapters 4 and 5 show how the ESS of a description, counted in demonstrations, differs from it.

Prior effective sample size. Prior ESS expresses the influence of a prior in observation units, with different definitions matching different statistical quantities (Morita et al., 2008; Neuenschwander et al., 2020). Our comparison is closest to measures based on expected loss (Reimherr et al., 2021; Wiesenfarth & Calderazzo, 2020). Le Scao & Rush (2021) measure the average horizontal separation of learning curves in prompt-based fine-tuning. Selby et al. (2025) compare language-model-elicited priors with data-based predictors by the number of observations needed to match predictive loss. We adapt performance matching to task descriptions at a specified demonstration count, using each learner's own baseline curve, and analyze how precision, dimension, and relevance probability determine predictive ESS.

These definitions extend the ESS beyond the conjugate case in different ways. Morita et al. (2008) extend it to any parametric prior. Their ESS is the number of hypothetical observations that, starting from a very vague prior, would give a posterior carrying about the same information as the prior, where information is measured by the curvature of the log density. Neuenschwander et al. (2020) require an ESS to be predictively consistent: on average, the ESS of the posterior after N observations should equal the prior's ESS plus N. Other definitions let the ESS depend on the current sample, through its size or through the observed data themselves. Reimherr

et al. (2021) compare a prior with a baseline prior. Their ESS is the number of observations that the baseline would need to perform as well at a given sample size, and it can be negative when the prior and the likelihood conflict. Similarly, the effective current sample size of Wiesenfarth & Calderazzo (2020) counts the observations that must be added to, or removed from, the current sample for an analysis without the prior to match the analysis with it. Finally, Reznik (2026) measures information by predictive log loss to choose the discount of a power prior. A power prior raises the likelihood of historical data, which may not match the current data, to a power between 0 and 1.

Our predictive ESS (Definition 3.1) is the number of additional demonstrations that a learner without the description needs to match the learner's error with it. Like the measure of Reznik (2026), it measures information by predictive loss, here squared error.

Our ESS also differs from that of Morita et al. (2008) in what it matches. Their ESS matches information. In the conjugate normal model above, a prior of precision c adds c to the posterior precision whatever the number of observations, so it is worth c observations at every sample size. In the same way, a description that raises the prior precision from a_0^{-1} to c in every direction is worth $c - a_0^{-1} = (1 - r)\, c$ by their measure, whatever the number of demonstrations in hand. Our ESS matches errors instead. In d dimensions, n demonstrations leave the Bayesian predictor an error of $d/(n - d - 1)$ under a flat prior (Section 2.4), not the d/n that their expected information would give. Matching errors therefore adds a term that grows with the dimension, the dimension term of Chapter 4. This term fades as demonstrations accumulate, and with many demonstrations in hand the ESS_n of a reliable description returns to their value (Corollary 5.2(ii)).

2.6 Robust Mixture Priors

A prior built from historical data can conflict with the current data. Evans & Moshonov (2006) check for such prior–data conflict by asking whether the observed data are surprising under the prior predictive distribution. Clinical trials that borrow historical control data protect against conflict with a robust mixture prior (Schmidli et al., 2014). The informative prior f_I, derived from the historical data, is mixed with a vague prior

f_V:

$$f(w) = p\, f_I(w) + (1 - p)\, f_V(w).$$

The posterior is then a mixture of the two component posteriors, and the weight on the informative component is updated in proportion to p times its marginal likelihood. When the data agree with the informative component, its weight grows, and the trial borrows the historical information. When they conflict, the weight moves to the vague component, and the analysis falls back on the current data.

A description is a prior of the same kind. It may describe the task at hand, but it may also have been written for another task, or it may simply be wrong. Chapter 3 therefore models a description as a robust mixture prior whose weight is its reliability p, the probability that the description is relevant. Chapter 5 calls the updated weight the posterior relevance probability π_n and shows how reliability affects the description's ESS.

2.7 Descriptions in Theories of ICL

Additional task information has been studied through input-distribution descriptions (Huang & Ge, 2025), candidate hypotheses (Lin et al., 2025), task-group mixtures (Lin & Lee, 2024), changing instruction influence (Tong et al., 2026), and related datasets supplied as context (Zhu et al., 2026). We use explicit Bayesian predictors to quantify how such information changes the demonstrations needed for a given prediction error.

In some of these theories, the description is a set of tokens that indicate a hypothesis class (Lin et al., 2025) or carry the mean of the inputs (Huang & Ge, 2025). These theories generally do not measure the value of a description in demonstrations, and they do not model the possibility that it is irrelevant. For example, Huang & Ge (2025) show that a learner can use the description token to predict better, but they do not say how many demonstrations it is worth or what it is worth when it may be wrong. The Bayesian theory of Xie et al. (2022) models the prompt as demonstrations alone. We keep its view of the prompt as evidence about a latent task and add a description, which induces a distribution over tasks before the demonstrations are observed. Our networks in Chapter 6 use the prefix embedding of Huang & Ge (2025), in which the description is a single token placed before the demonstrations. The work closest to our setting is that of Zhu et al. (2026), who show that a Transformer given

a prefix of related datasets performs amortized hierarchical Bayesian prediction. We measure the value of such conditioning information in demonstrations, and we allow it to be irrelevant.

2.8 Descriptions and Priors in Pretrained Language Models

Task summaries distilled from demonstrations can replace the examples in context (Honda et al., 2025). Other studies report weak sensitivity to instruction relevance (Webson & Pavlick, 2022) or limited benefits from repository instruction files (Gloaguen et al., 2026). Gupta et al. (2025) compare explicitly stated coin biases with observations and evaluate the predictions against Bayesian references. Concurrent work studies how reasoning and code execution affect language models' approximation of Bayesian posteriors (Filbry et al., 2026). Our numerical experiments vary stated precision and reliability separately, with controls that hold the description center and demonstrations fixed. Code execution is disabled in the main runs. Section 2.9 discusses further connections.

2.9 Additional Related Work

Task descriptions and demonstrations. Weller et al. (2020) formalize learning from task descriptions by distinguishing the latent task, its natural-language description, and individual input–output examples. We use *task description* for information about the task supplied in the prompt, including a numeric description token in the meta-trained setting and probabilistic statements or policy constraints in the language-model experiments. A *demonstration* is an observed input–output example; in scalar mean estimation, it is an observation at the fixed input $x = 1$. Summaries distilled from demonstrations can serve as descriptions (Honda et al., 2025); our controlled experiments instead specify the joint distribution of task, description, and demonstrations.

Prior information and uncertain relevance. The two-component construction is related to robust mixture priors, which combine informative and baseline distributions

and update their component weights after observing data (Schmidli et al., 2014). Our mixture is specified directly, rather than derived from a historical-data meta-analysis. Reliability p is the prior probability of the description-conditioned component; π_n is its posterior probability. Averaging component predictions using π_n is Bayesian model averaging (Hoeting et al., 1999). The experimental comparison that retains p instead averages with prior weights. Fixed prior weights in a robust prior still yield updated posterior weights.

Our relevance terminology also has a conceptual precedent in information fusion, where relevance is distinguished from truthfulness (Pichon et al., 2012). That work uses belief functions; our probability model is different. In our model, an irrelevant description is independent of the task and may happen to agree with it. Thus irrelevance is a generating condition, whereas prior–data conflict concerns disagreement between prior information and observed data. Related approaches discount external information according to predictive loss (Reznik, 2026) or accommodate conflict with heavy-tailed distributions (O'Hagan & Pericchi, 2012).

Specificity and description weights. Specificity takes a concrete form in each setting. In the isotropic Gaussian model, it is represented by prior precision c; a partial description also specifies the number k of constrained directions. In the semantic experiment, broad and specific notes leave different numbers of compatible policies. These constructions do not define a common numerical scale across tasks. In Gaussian mean estimation, the reference combines the prior mean and sample mean with weights proportional to their precisions (Neuenschwander et al., 2020). Fitted and per-response description weights summarize model predictions. Their normalized values can be compared with posterior component probabilities, but need not themselves be probabilities.

Effective sample size and predictive comparisons. Prior ESS has several definitions, including matching local information (Morita et al., 2008), matching expected posterior loss (Reimherr et al., 2021), and quantifying sample savings at a target estimation accuracy (Wiesenfarth & Calderazzo, 2020). Our definition follows the risk-matching approach but evaluates population prediction error and allows a non-Bayesian learner. Its dependence on the demonstration count is part of the definition. The word *predictive* specifies what is matched; it does not assert the

additive predictive-consistency property studied by Neuenschwander et al. (2020).

Two predictive comparisons are especially close. Le Scao & Rush (2021) average horizontal data advantages over an accuracy range in supervised fine-tuning; we evaluate a pointwise difference in demonstration counts. Selby et al. (2025) compare priors elicited from language models with data-based predictors by the observations needed to match predictive loss; we study descriptions supplied to a learner and how their benefit changes after demonstrations. Beyond ICL, an instruction can replace up to 100 execution examples in the fine-tuning settings studied by Cook et al. (2026). Wenliang et al. (2026) compare optimized prompts with demonstrations in terms of their length and effect. They treat a prompt as conditioning for a Bayesian sequence predictor and find that an optimized prompt can be shorter than demonstrations with the same effect.

Demonstrations, text, and probabilistic prediction. Other work examines semantic priors and flipped labels (Wei et al., 2023), weak dependence on demonstration labels (Min et al., 2022), Bayesian scaling laws (Arora et al., 2025), and Bayesian reference curves for in-context function learning (Akata et al., 2026). Text supplied alongside numbers can change numerical predictions (Requeima et al., 2024), and models estimate a density from numbers in context like a kernel density estimator with an adaptive bandwidth (Liu et al., 2025). Related work studies how language-model outputs compare with Bayesian inference and how training can improve probabilistic reasoning (Qiu et al., 2026); He et al. (2025) examine Bayesian consistency during iterative reasoning on open-ended questions. Our numerical experiments state uncertainty explicitly, allowing comparison with a specified Bayesian reference as precision and relevance probability vary. The semantic task extends this comparison to ordinary-language constraints on a finite family of classification policies.

In the reference comparisons, the learning curves of pretrained models approach a Bayesian reference on synthetic tasks with a known optimum (Akata et al., 2026), and Arora et al. (2025) model in-context learning curves with a Bayesian scaling law that includes a task prior. Where Min et al. (2022) and Wei et al. (2023) vary the labels of the demonstrations, we ask the converse question: does a model weigh a description according to what the prompt says about it?

Chapter 3

Framework

Tasks and demonstrations. We use the standard noisy in-context linear regression setting (Garg et al., 2022; Raventós et al., 2023). A task is a vector $w \in \mathbb{R}^d$ with baseline prior $\mathcal{N}(0, a_0 I_d)$, where $a_0 > 0$. A prompt contains n demonstrations $D_n = \{(x_i, y_i)\}_{i=1}^n$ and a query input x, with

$$x_i, x \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d), \qquad y_i = w^\top x_i + \epsilon_i, \quad \epsilon_i \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

Inputs and observation noise are independent of the task and one another. The parameter a_0 controls variation across tasks. Numerical illustrations in Chapters 4–5 use $d = 16$ and $a_0 = 10$.

Descriptions and relevance. A description supplies a center $m \in \mathbb{R}^d$. Its *specificity* is represented by precision $c \geq a_0^{-1}$, and its *reliability* is $p \in [0, 1]$. Let $r = (ca_0)^{-1} \in (0, 1]$ be the fraction of baseline variance remaining in the relevant component. With a latent relevance indicator $Z \sim \text{Bernoulli}(p)$ independent of m, the joint model is

$$\begin{aligned} m &\sim \mathcal{N}(0, (1-r)a_0 I_d), \\ w \mid m, Z = 1 &\sim \mathcal{N}(m, r a_0 I_d), \\ w \mid m, Z = 0 &\sim \mathcal{N}(0, a_0 I_d). \end{aligned}$$

Thus

$$w \mid m \sim p\,\mathcal{N}(m, c^{-1} I_d) + (1-p)\,\mathcal{N}(0, a_0 I_d).$$

This two-component structure is used in robust mixture priors (Schmidli et al., 2014). Integrating over m recovers $w \sim \mathcal{N}(0, a_0 I_d)$ under either component, so comparisons with and without descriptions concern the same task population.

A description is *relevant* when $Z = 1$ and *irrelevant* when $Z = 0$. Relevance identifies the generating component, irrespective of the realized distance between m and w. We use *reliable* for $p = 1$ and *unreliable* for $p < 1$. At $c = a_0^{-1}$, the two components coincide and the description provides no task information. The model separates uncertainty within the described region from uncertainty about whether that region applies. It does not assume that every natural-language description can be summarized by these two scalars.

A relevant description splits the baseline variance a_0 into the part $(1 - r)a_0$ that m resolves and the part ra_0 that it leaves. Because the observation noise has unit variance, c is measured in demonstrations: in each direction, it is the ESS that Morita et al. (2008) assign to a normal prior. We call a description *loose* when c is small relative to d and *specific* when c exceeds d. At $d = 16$ and $a_0 = 10$, a description with $c = 1$ ($r = 0.1$) is loose, and one with $c = 100$ ($r = 0.001$) is specific.

Prediction error. For a prediction $\hat{y}$, we measure expected squared error against the noise-free query response:

$$R = \mathbb{E}\big[(\hat{y} - w^\top x)^2\big]. \tag{3.1}$$

We call this quantity *prediction error*, or *risk*; it excludes irreducible query noise. The expectation is over tasks, descriptions, demonstrations, and queries. For a fixed learner, $R^{\text{desc}}(n)$ and $R^{\text{demo}}(n)$ denote risk with and without a description, and $R^{\text{desc}}_{p=1}(n)$ denotes the reliable case.

Definition 3.1 (Predictive effective sample size)**.** Given n demonstrations, let n^* be the smallest $t \geq 0$ satisfying $R^{\text{demo}}(t) = R^{\text{desc}}(n)$. When this crossing exists, the description's ESS is

$$\text{ESS}_n = n^* - n.$$

Noninteger counts refer to an interpolated baseline curve.

We interpolate risk linearly in demonstration count for the theory, meta-trained Transformers, and semantic task, and linearly in log count for numerical language-model tasks. Empirical ESS uses the first crossing in the evaluated range. When no crossing is observed, we report censoring at the corresponding boundary under the conventions in Section 7.7; an exact ESS is then unavailable.

An ESS of 10 means that n demonstrations plus the description match the error of $n + 10$ demonstrations alone. For a decreasing baseline curve, negative ESS indicates higher error at the same demonstration count. Because each learner uses its own baseline, a large ESS need not imply low absolute error. For the Bayes-optimal predictor under the correctly specified model, conditioning on descriptions cannot increase expected risk, so ESS is nonnegative.

ESS thus measures a horizontal displacement of a learning curve, and the description's value *to that learner*. Because R averages over descriptions, the ESS belongs to a kind of description, determined by c and p, rather than to a particular value of m.

Chapter 4

The Value of a Reliable Description

For $p = 1$, a description with fixed positive integer precision c has $\mathrm{ESS}_0 \to c + d + 1$ as $a_0 \to \infty$. Its value without demonstrations therefore depends on task dimension as well as precision. The Bayes-optimal predictor uses the posterior mean of $w^\top x$, so its risk is the expected trace of the posterior covariance of w. Write $W_n = X_n^\top X_n$, where the rows of X_n are the demonstration inputs. Gaussian conjugacy gives the Bayes-optimal risks

$$R^{\mathrm{demo}}(n) = \mathbb{E}\,\mathrm{tr}(a_0^{-1} I_d + W_n)^{-1},$$
$$R^{\mathrm{desc}}_{p=1}(n) = \mathbb{E}\,\mathrm{tr}(c I_d + W_n)^{-1}.$$

The description replaces baseline precision a_0^{-1} with c. Its predictive value is not, however, determined by this precision difference alone.

Proposition 4.1 (Reliable descriptions)**.** *Let $p = 1$.*

(i) With $r \in (0, 1]$ fixed, $\mathrm{ESS}_0 \geq (1 - r)d$, and $\mathrm{ESS}_0 \to (1 - r)d$ as $a_0 \to \infty$.

(ii) With positive integer c fixed, $\mathrm{ESS}_0 \to c + d + 1$ as $a_0 \to \infty$.

The fixed-precision limit in part (ii) follows by comparing the description's risk d/c with the limiting demonstrations-only risk $d/(n - d - 1)$, valid for integer $n > d + 1$ (Muirhead, 1982). The risks match at $n = c + d + 1$. Integer c places this crossing on the grid used to interpolate the learning curve. In this limit, demonstrations-only risk depends on $\mathbb{E}\,\mathrm{tr}(W_n^{-1}) = d/(n - d - 1)$, whereas $\mathbb{E}W_n = nI_d$; this difference accounts for the additional $d + 1$ demonstrations.

This $d + 1$ contribution reflects the cost of learning a d-dimensional task from random inputs. Matching average information would count only the added precision

$c - a_0^{-1}$; matching predictive error also accounts for the random design. We refer to the additional contribution as the *dimension term*. It is substantial when c is small relative to d, whereas precision dominates when c is large. This interpretation concerns the stated limits, rather than an exact additive formula for every finite a_0 and n.

The comparison with matching information can be made explicit in this limit. For a normal prior, the ESS of Morita et al. (2008) is the ratio of the observation variance to the prior variance, so raising the baseline precision a_0^{-1} to c is worth $c - a_0^{-1} = (1 - r)c$ demonstrations under their definition, at every n. With a diffuse baseline prior, n demonstrations leave risk $d/(n - d - 1)$, not the risk d/n that their expected information would suggest. Setting d/n equal to the description's risk d/c gives $n = c$; setting $d/(n - d - 1)$ equal to it gives $n = c + d + 1$. The dimension term is the gap between the two.

Part (i) describes a different scaling. Holding r fixed makes $c = (ra_0)^{-1}$ decrease as a_0 grows, so the description removes a fixed fraction $1 - r$ of baseline variance. Before the demonstration inputs span $\mathbb{R}^d$, error in the remaining directions dominates as $a_0 \to \infty$. The description then replaces $(1 - r)d$ demonstrations in the limit, with the same quantity providing a lower bound at finite a_0 (Section 4.1).

At $d = 16$ and $a_0 = 10$, a reliable description with $c = 1$ has ESS 15.6, compared with the lower bound $(1 - r)d = 14.4$. Increasing c to 100 raises ESS to approximately 117 (Figure 4.1). With many demonstrations, the dimension-dependent contribution disappears: $\mathrm{ESS}_n \to c - a_0^{-1}$ (Corollary 5.2). After 100 demonstrations, the $c = 1$ description has ESS 1.1, close to its added precision of 0.9.

The number of constrained directions matters. A reliable description may constrain only $k < d$ directions. With fixed $c > 0$, its zero-demonstration ESS approaches k as $a_0 \to \infty$: the unconstrained directions dominate the remaining error. With many demonstrations, its ESS instead approaches $k(c - a_0^{-1})/d$ (Proposition 4.2, Section 4.2). A sufficiently precise partial description can therefore gain value as demonstrations resolve what it leaves unspecified. At $d = 16$, $k = 8$, and $c = 100$, ESS rises from 8.1 without demonstrations to 36.2 after 100 demonstrations (Table 4.1).

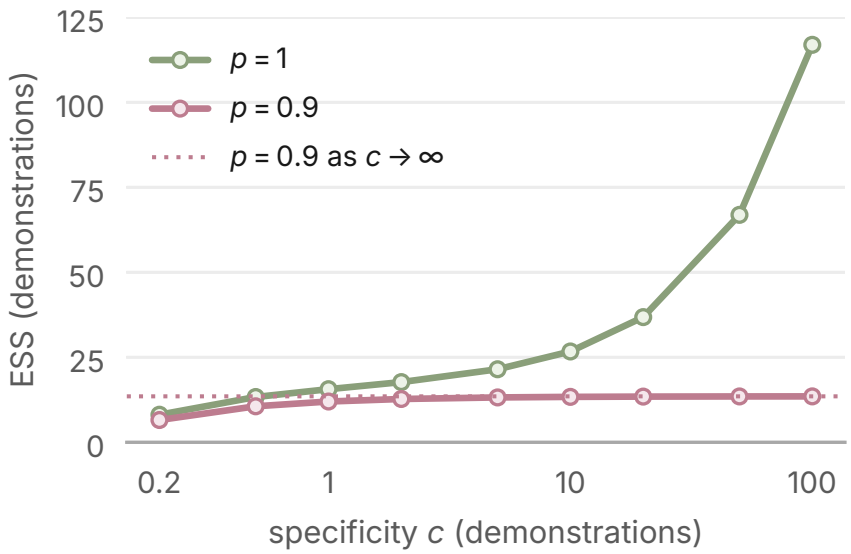


Figure 4.1: **Reliability limits the benefit of specificity without demonstrations.** Bayes-optimal ESS_0 versus precision c, for $p = 1$ and $p = 0.9$, with $d = 16$ and $a_0 = 10$. The dotted line is the finite ceiling as $c \to \infty$ at $p = 0.9$. Estimates use eight batches of 4,000 prompts; standard errors are below 0.02 demonstrations.

4.1 Reliable Descriptions

Notation and risk. Write $\varepsilon = 1/a_0$ for the baseline prior precision. Let $X_n \in \mathbb{R}^{n \times d}$ contain the demonstration inputs as rows, and set $W_n = X_n^\top X_n$. For a prior precision $\kappa > 0$, define

$$R^\kappa(n) = \mathbb{E}\,\mathrm{tr}(\kappa I_d + W_n)^{-1}.$$

Thus $R^{\mathrm{demo}} = R^\varepsilon$ and $R^{\mathrm{desc}}_{p=1} = R^c$. Here and throughout, risk means expected squared prediction error against the noiseless query mean $w^\top x$. The identities follow from the Gaussian posterior covariance: the posterior mean minimizes conditional squared error, and the independent query satisfies $\mathbb{E}[xx^\top] = I_d$. Averaging the conditional error $x^\top \Sigma x$ therefore gives $\mathbb{E}\,\mathrm{tr}\,\Sigma$.

Throughout the proofs, d is a positive integer, $a_0 > 0$, $c \geq \varepsilon$, $r = 1/(ca_0) \in (0, 1]$, and $p \in [0, 1]$. Specificity is represented by the relevant component's precision c; r is its variance as a fraction of the baseline variance. Reliability p is the prior probability of relevance. Noninteger demonstration counts refer to linear interpolation between the risks at adjacent integers.

Table 4.1: **Demonstrations can increase the value of a precise partial description.** Bayes-optimal ESS for a reliable description of k of $d = 16$ directions, with $a_0 = 10$. The case $k = 16$ is a complete description. Each block reports ESS after n demonstrations and its large-sample limit $(1 - r)ck/d$. Estimates use eight batches of 2,000 prompts; standard errors are at most 0.05.

	$c = 1$					$c = 100$				
k	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$	$n = 0$	$n = 10$	$n = 100$	$n = 400$	$n \to \infty$
1	0.9	0.8	0.1	0.1	0.1	1.0	1.0	3.5	5.2	6.2
4	3.6	2.9	0.3	0.2	0.2	4.0	4.2	15.5	21.7	25.0
8	7.3	5.1	0.5	0.5	0.5	8.1	9.8	36.2	46.0	50.0
12	11.1	6.7	0.8	0.7	0.7	12.4	23.4	65.1	73.0	74.9
16	15.6	8.0	1.1	0.9	0.9	117	116	109	103	99.9

Wishart identities. We use three standard properties of W_n.

(W1) If $n < d$, then W_n has rank n almost surely. Consequently, $(\kappa I_d + W_n)^{-1}$ has $d - n$ eigenvalues equal to $1/\kappa$.

(W2) If $n \geq d$ and e is a fixed unit vector, then $(e^\top W_n^{-1} e)^{-1} \sim \chi^2_{n-d+1}$ (Muirhead, 1982, Theorem 3.2.12). Hence

$$\mathbb{E}\,\mathrm{tr}\, W_n^{-1} = \frac{d}{n - d - 1} \quad (n \geq d + 2),$$

whereas this expectation is infinite for $n = d$ and $n = d + 1$.

(W3) For fixed d, as $n \to \infty$,

$$\mathbb{E}\,\mathrm{tr}\, W_n^{-2} = \frac{d}{n^2} + O(n^{-3}), \qquad \mathbb{E}\,\mathrm{tr}\, W_n^{-3} = O(n^{-3})$$

(von Rosen, 1988).

The Sherman–Morrison identity implies that an additional demonstration strictly decreases the expected trace of the posterior covariance. Thus $R^\kappa(n)$ is strictly decreasing, and (W2) implies that it tends to zero. Its linear interpolation is therefore continuous and strictly decreasing. In the correctly specified joint model, conditioning on the description cannot increase expected posterior variance, so $R^{\mathrm{desc}}(n) \leq R^{\mathrm{demo}}(n)$ and the Bayes-optimal ESS is nonnegative.

Proof of Proposition 4.1. A reliable description without demonstrations has risk

$$R^{\mathrm{desc}}_{p=1}(0) = \frac{d}{c} = ra_0 d.$$

(i) Fixed variance reduction. By (W1), $R^{\varepsilon}(n) \geq (d-n)a_0$ for $n < d$; the inequality also holds trivially for $n \geq d$. Linear interpolation preserves this bound, so

$$R^{\mathrm{demo}}(t) \geq (d-t)a_0 \qquad (t \geq 0).$$

For $t < (1-r)d$, the right-hand side exceeds $ra_0 d$. The matching sample size must therefore satisfy $\mathrm{ESS}_0 \geq (1-r)d$.

For every integer n,

$$\frac{R^{\varepsilon}(n)}{a_0} = \mathbb{E}\,\mathrm{tr}(I_d + a_0 W_n)^{-1} \longrightarrow (d-n)^+ \qquad \text{as } a_0 \to \infty.$$

Indeed, an eigenvalue $1/(1+a_0\lambda)$ is bounded by one and converges to the indicator of $\lambda = 0$, so dominated convergence and the almost-sure rank of W_n apply. The same limit holds at every real t by interpolation. For fixed $r \in (0,1)$, the limiting curve crosses rd at the unique point $t = (1-r)d$. Strict monotonicity then gives convergence of the crossing. For $r = 1$, the description is uninformative and $\mathrm{ESS}_0 = 0$ for every a_0, giving the same conclusion.

(ii) Fixed precision. For fixed $c > 0$, the description risk d/c is independent of a_0. As $a_0 \to \infty$, monotone convergence, (W1), and (W2) give

$$R^{\varepsilon}(n) \longrightarrow \begin{cases} +\infty, & n \leq d+1, \\ \dfrac{d}{n-d-1}, & n \geq d+2. \end{cases}$$

If c is a positive integer, the finite part of this limiting curve equals d/c at the integer $n = c+d+1$. For every fixed point below this crossing the limiting risk is larger than d/c, and for every fixed point above it the limiting risk is smaller. The crossing of the interpolated finite-a_0 risk therefore converges to $c+d+1$. The integer condition ensures that the equality occurs at a grid point; linear interpolation does not in general preserve the same expression for noninteger c. □

Large-sample expansion. For fixed d and κ, commuting W_n with κI_d gives

$$(\kappa I_d + W_n)^{-1} = W_n^{-1} - \kappa W_n^{-2} + \kappa^2 W_n^{-2}(\kappa I_d + W_n)^{-1}.$$

The final term is positive semidefinite and bounded above by $\kappa^2 W_n^{-3}$. Using (W2) and (W3),

$$R^\kappa(n) = \frac{d}{n-d-1} - \frac{\kappa d}{n^2} + O(n^{-3}). \tag{4.1}$$

An additional demonstration reduces the leading term by $d/n^2 + O(n^{-3})$. A fixed increase in prior precision and a fixed increase in the demonstration count therefore reduce risk at the same asymptotic order. This comparison gives the large-sample ESS limits below.

4.2 Partial Descriptions

A reliable description may constrain only $k < d$ directions. Suppose the description leaves a fraction r of the baseline variance in these directions and leaves the remaining $d-k$ directions unchanged. The description-conditioned prior precision is c in the constrained directions and $\varepsilon = 1/a_0$ elsewhere. Write

$$\Lambda = \operatorname{diag}(cI_k, \varepsilon I_{d-k}).$$

The prediction risk is $\mathbb{E}\,\operatorname{tr}(\Lambda + W_n)^{-1}$. Rotational invariance of W_n means that the result depends on the number of constrained directions, not on their orientation.

Proposition 4.2 (ESS of a partial description)**.** *Let $p = 1$ and let $0 \le k < d$ be an integer.*

(i) As $a_0 \to \infty$, $\mathrm{ESS}_0 \to (1-r)k$ with r fixed, and $\mathrm{ESS}_0 \to k$ with c fixed.

(ii) For fixed d, a_0, and c, as $n \to \infty$,

$$\mathrm{ESS}_n \longrightarrow \frac{k}{d}(c - 1/a_0) = \frac{k}{d}(1-r)c.$$

As the baseline prior becomes diffuse with c fixed, the unconstrained directions dominate the zero-demonstration risk, and ESS approaches the number of constrained directions, k. After many demonstrations, ESS instead approaches the added precision averaged across all d directions. A sufficiently precise partial description can therefore be worth more alongside demonstrations than by itself. Table 4.1 illustrates this complementarity: at $c = 100$, a description of half of the $d = 16$ directions has ESS 8.1 without demonstrations and 36 with 100 demonstrations.

The large-sample limit exceeds k about when $c - a_0^{-1} > d$, and this is when a partial description gains value as demonstrations accumulate. Between 0 and

100 demonstrations, a complete description at $c = 100$ falls from 117 to 109, so demonstrations substitute for a complete description and complement a partial one. At $c = 1$, ESS falls as demonstrations accumulate, whether the description is partial or complete.

Proof of Proposition 4.2. (i) No demonstrations. The description risk is

$$\operatorname{tr} \Lambda^{-1} = (kr + d - k)a_0.$$

Using $R^{\mathrm{demo}}(t) \geq (d - t)a_0$ gives $\mathrm{ESS}_0 \geq (1 - r)k$. The normalized baseline risk converges to $(d-t)^+$, locally uniformly in t, by the integer limits and linear interpolation. With r fixed, the normalized description risk is $d - (1 - r)k$. With c fixed, it is

$$d - k + \frac{k}{ca_0} \longrightarrow d - k.$$

Because $k < d$, both limiting levels are positive and have unique crossings, at $(1 - r)k$ and k, respectively. Convergence of the crossings proves the claim. When the description leaves the prior unchanged, its ESS is identically zero.

(ii) Many demonstrations. Applying the resolvent identity twice gives

$$(\Lambda + W_n)^{-1} = W_n^{-1} - W_n^{-1}\Lambda W_n^{-1} + W_n^{-1}\Lambda W_n^{-1}\Lambda(\Lambda + W_n)^{-1}.$$

Rotational invariance implies

$$\mathbb{E} W_n^{-2} = \frac{\mathbb{E} \operatorname{tr} W_n^{-2}}{d} I_d.$$

The expected trace of the middle term is therefore $\operatorname{tr}(\Lambda)/n^2 + O(n^{-3})$ by (W3). To bound the remainder, use $|\operatorname{tr}(MB)| \leq \|M\|_{\mathrm{op}} \operatorname{tr} B$ for $B \succeq 0$, together with $(\Lambda + W_n)^{-1} \preceq W_n^{-1}$. Its absolute trace is at most

$$\|\Lambda\|_{\mathrm{op}}^2 \|W_n^{-1}\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-1} \leq d\|\Lambda\|_{\mathrm{op}}^2 \operatorname{tr} W_n^{-3},$$

whose expectation is $O(n^{-3})$. Consequently,

$$\mathbb{E} \operatorname{tr}(\Lambda + W_n)^{-1} = \frac{d}{n - d - 1} - \frac{\operatorname{tr} \Lambda}{n^2} + O(n^{-3}).$$

Matching this expansion to the baseline risk, as in Corollary 5.2(ii), gives

$$\mathrm{ESS}_n \longrightarrow \frac{\operatorname{tr} \Lambda - d\varepsilon}{d} = \frac{k(c - \varepsilon)}{d}.$$

□

The same expansion applies to any fixed Gaussian description precision matrix $\Lambda \succeq \varepsilon I_d$ compatible with the baseline task distribution. Its large-sample ESS is the added precision averaged across directions, $(\operatorname{tr}\Lambda - d\varepsilon)/d$.

Chapter 5

The Cost of Unreliability

Uncertain relevance limits the benefit of increasing precision. For every fixed $p < 1$, the zero-demonstration risk remains positive as $c \to \infty$, giving a finite ESS ceiling.

For $0 < p < 1$, let $\mu_z = \mathbb{E}[w \mid m, D_n, Z = z]$ be the component posterior means and let $\pi_n = P(Z = 1 \mid m, D_n)$ be the *posterior relevance probability*. Bayesian model averaging (Hoeting et al., 1999) gives

$$\hat{y} = \big[\pi_n \mu_1 + (1 - \pi_n)\mu_0\big]^\top x.$$

Demonstrations affect both the component predictions and their relative weight. The law of total variance then yields the following risk decomposition.

Proposition 5.1 (Risk under uncertain relevance)**.** *For every $n \geq 0$,*

$$\begin{aligned} R^{\text{desc}}(n) = {} & p R^{\text{desc}}_{p=1}(n) \\ & + (1 - p) R^{\text{demo}}(n) + \mathcal{V}_n, \end{aligned} \tag{5.1}$$

where

$$\mathcal{V}_n = \mathbb{E}\Big[\pi_n(1 - \pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2\Big] \geq 0.$$

Without demonstrations,

$$R^{\text{desc}}(0) = \big[1 - p^2(1 - r)\big] a_0 d. \tag{5.2}$$

Here $\mathcal{V}_n$ is the expected between-component posterior variance. The first two terms in Equation (5.1) are the risk of the Bayes-optimal predictor that also observes Z, so $\mathcal{V}_n$ is the excess Bayes risk due to unobserved relevance. This oracle uses the description

when it is relevant and otherwise uses the baseline prior; $\mathcal{V}_n$ is the disagreement between these two predictions, weighted by uncertainty about which is appropriate. Without demonstrations, observing Z would remove a fraction $p(1-r)$ of baseline risk; without Z, the reduction is only $p^2(1-r)$ (Section 5.1).

The cost of uncertainty about relevance is small when the two predictions agree. It is also small when the demonstrations make one relevance branch far more likely than the other, because $\pi_n(1-\pi_n)$ is then near zero. Without demonstrations, the cost is $\mathcal{V}_0 = p(1-p)(1-r)a_0 d$, which grows as a more specific description pulls the two predictions apart. With demonstrations, however, a specific description is also easier to distinguish from an irrelevant one.

Corollary 5.2 (ESS under uncertain relevance)**.**

(i) For $0 < p < 1$, define $n_p^\sharp$ by

$$R^{\mathrm{demo}}(n_p^\sharp) = (1-p^2)R^{\mathrm{demo}}(0).$$

Then $\mathrm{ESS}_0 < n_p^\sharp$ at every finite c, and $\mathrm{ESS}_0 \to n_p^\sharp$ as $c \to \infty$. As $a_0 \to \infty$ with r fixed, $\mathrm{ESS}_0 \to p^2(1-r)d$ and $n_p^\sharp \to p^2 d$. For $p = 0$, both ESS and the ceiling are zero.

(ii) With d, a_0, and c fixed, as $n \to \infty$,

$$\begin{cases} \mathrm{ESS}_n \to c - a_0^{-1}, & p = 1, \\ \limsup_{n\to\infty} \mathrm{ESS}_n \le p(c - a_0^{-1}), & p < 1. \end{cases}$$

Uncertain relevance imposes an error floor without demonstrations. For fixed $p < 1$, Equation (5.2) approaches the positive floor $(1-p^2)a_0 d$ as $c \to \infty$, giving a finite ceiling on zero-demonstration ESS. At $p = 0.9$, $d = 16$, and $a_0 = 10$, increasing c from 1 to 100 raises ESS only from 12.0 to 13.5, compared with 15.6 to 117 at $p = 1$ (Figure 4.1). The finite-a_0 ceiling is approximately 13.5; its limit as the baseline prior becomes diffuse is $p^2 d = 12.96$.

The oracle that observes Z has the lower floor $(1-p)a_0 d$, which is the risk of a reliable description that leaves variance fraction $1-p$. In this setting, its ceiling is therefore 15.6 demonstrations, the ESS of the reliable $c = 1$ description; uncertainty about relevance lowers it to approximately 13.5.

With many demonstrations, ESS is at most p times that of a reliable description. Demonstrations reduce the risk when the description is irrelevant, $R^{\text{demo}}(n)$, so the term $(1-p)R^{\text{demo}}(n)$ falls and ESS can grow. In the limit, a reliable description is worth $c - a_0^{-1}$ demonstrations: the dimension benefit has disappeared, and what remains is the precision the description adds to the baseline prior. An unreliable description is then worth at most p times as much (Corollary 5.2). After 100 demonstrations, the $c = 100$ description has ESS 109 when $p = 1$ and 86.6 when $p = 0.9$, compared with limits of 99.9 and at most 89.9. The reliable value exceeds its limit because the dimension benefit has not yet disappeared completely at $n = 100$.

Demonstrations distinguish low precision from uncertain relevance. Without demonstrations, an unreliable description has the same risk as a reliable description with remaining variance fraction $1 - p^2(1-r)$. For example, $c = 100$, $p = 0.9$ and a reliable description with $c \approx 0.52$ both give ESS about 13.5 here. After 100 demonstrations, their ESS values are 86.6 and 0.5. Demonstrations can support the high-precision component of the unreliable description; the reliable low-precision description has no such component. A predictor that assumes relevance by fixing $p = 1$ can instead increase error (Section 5.2).

Assuming relevance can be harmful. The Bayesian predictor assuming relevance always uses μ_1, even when the description is irrelevant. Without demonstrations, it is worse than ignoring the description when $p < 1/2$, provided the description is informative ($r < 1$). For fixed n, its risk tends to $2(1-p)a_0 d$ as $c \to \infty$, whereas the demonstrations-only risk tends to zero as n grows. Thus a sufficiently precise, occasionally irrelevant description can outweigh useful demonstrations. At $p = 0.9$ and $c = 100$, the predictor assuming relevance has higher risk than the demonstrations-only predictor from 15 demonstrations onward, at every count evaluated up to 10^5 (Section 5.2).

5.1 Unreliable Descriptions

Let $\mathcal{P}_n = (D_n, m, x)$ denote the full prompt, including the query input. The posterior relevance probability is $\pi_n = P(Z = 1 \mid \mathcal{P}_n)$, whereas p remains the prior relevance

probability. Conditional on $Z = 1$, the posterior parameters are

$$\Sigma_1 = (cI_d + W_n)^{-1}, \qquad \mu_1 = \Sigma_1(cm + X_n^\top y_{1:n}).$$

Conditional on $Z = 0$, the description center m is independent of w, and

$$\Sigma_0 = (\varepsilon I_d + W_n)^{-1}, \qquad \mu_0 = \Sigma_0 X_n^\top y_{1:n}.$$

The posterior relevance probability follows from the two Gaussian marginal likelihoods,

$$\begin{aligned} L_1 &= \mathcal{N}\big(y_{1:n}; X_n m, I_n + c^{-1} X_n X_n^\top\big)\,, \\ L_0 &= \mathcal{N}\big(y_{1:n}; 0, I_n + a_0 X_n X_n^\top\big)\,, \\ \pi_n &= \frac{pL_1}{pL_1 + (1-p)L_0}. \end{aligned}$$

The query input does not enter this update because it is independent of (Z, m, D_n). Without demonstrations, both marginal likelihoods equal one and $\pi_0 = p$, since m has the same distribution under both relevance states. At $p = 0$ or $p = 1$, the posterior contains only the corresponding component. At $r = 1$, $m = 0$ almost surely and the two components coincide, so the description provides no additional task information for any p.

Proof of Proposition 5.1. As in Bayesian model averaging (Hoeting et al., 1999), the posterior mean prediction averages the component predictions using their posterior probabilities:

$$\mathbb{E}[w^\top x \mid \mathcal{P}_n] = \big(\pi_n \mu_1 + (1-\pi_n)\mu_0\big)^\top x.$$

The law of total variance over Z separates posterior variance into within-component and between-component terms:

$$\begin{aligned} \mathrm{Var}(w^\top x \mid \mathcal{P}_n) = \pi_n x^\top \Sigma_1 x + (1-\pi_n) x^\top \Sigma_0 x \\ + \pi_n(1-\pi_n)\big((\mu_1 - \mu_0)^\top x\big)^2. \end{aligned}$$

For $0 < p < 1$ and any integrable function g of the prompt,

$$\mathbb{E}[\pi_n g(\mathcal{P}_n)] = \mathbb{E}[Z g(\mathcal{P}_n)] = p\,\mathbb{E}[g(\mathcal{P}_n) \mid Z = 1].$$

Conditional on $Z = 1$, the prompt follows the reliable-description model. Conditional on $Z = 0$, the description provides no information about the task. Taking expectations in the variance decomposition therefore gives

$$R^{\mathrm{desc}}(n) = pR^{\mathrm{desc}}_{p=1}(n) + (1-p)R^{\mathrm{demo}}(n) + \mathcal{V}_n,$$

with $\mathcal{V}_n$ as in Equation (5.1). The endpoint cases $p = 0$ and $p = 1$ follow directly from the single-component posteriors.

At $n = 0$, $\mu_1 = m$, $\mu_0 = 0$, and the prediction is $p\, m^\top x$. For each coordinate,

$$\mathrm{Var}(w_i) = a_0, \qquad \mathrm{Var}(m_i) = (1-r)a_0, \qquad \mathrm{Cov}(w_i, m_i) = p(1-r)a_0.$$

The covariance holds because, conditional on relevance, w_i is m_i plus independent noise, and conditional on irrelevance, w_i is independent of m_i. It follows that

$$\mathbb{E}(w_i - pm_i)^2 = a_0 - 2p^2(1-r)a_0 + p^2(1-r)a_0 = \big(1 - p^2(1-r)\big)a_0.$$

Since $\mathbb{E}[xx^\top] = I_d$, the prediction risk is $\mathbb{E}\|w - pm\|^2$. Summing over coordinates proves Equation (5.2). □

Proof of Corollary 5.2. (i) No demonstrations. For $0 < p < 1$ and finite c,

$$R^{\mathrm{desc}}(0) = \big(1 - p^2(1-r)\big)R^{\mathrm{demo}}(0) > (1-p^2)R^{\mathrm{demo}}(0).$$

The risk on the left decreases to the right-hand side as $c \to \infty$, equivalently $r \to 0$. The interpolated baseline risk is continuous, strictly decreasing, and tends to zero. Consequently, the matching sample size is strictly below the unique finite $n_p^\sharp$ defined in the corollary and converges to it as $c \to \infty$. At $p = 0$, the description is always irrelevant, so $\mathrm{ESS}_0 = n_0^\sharp = 0$; the non-strict bound remains valid.

The baseline lower bound and normalized limit established in Section 4.1 also give

$$\mathrm{ESS}_0 \geq p^2(1-r)d, \qquad \mathrm{ESS}_0 \longrightarrow p^2(1-r)d$$

when $a_0 \to \infty$ with r fixed. Applying the same argument to the level $1 - p^2$ gives $n_p^\sharp \to p^2 d$. These expressions are limits as the baseline prior becomes diffuse; they need not hold exactly at finite a_0.

For comparison, an oracle predictor that observes Z has zero-demonstration risk

$$\big(1 - p(1-r)\big)a_0 d \geq (1-p)a_0 d.$$

The oracle's zero-demonstration ESS ceiling therefore approaches pd as $a_0 \to \infty$. For the learner that must infer relevance, this limit is $p^2 d$. The difference is the cost of uncertainty about relevance.

(ii) Many demonstrations. The idea is as follows. With many demonstrations, an additional demonstration reduces risk by about d/n^2, and a prior of precision κ

reduces it by κ times as much (Equation (4.1)). A reliable description raises the precision from ε to c, so it saves about $c - \varepsilon$ demonstrations. A description that is relevant with probability p saves about p times as much, and the term $\mathcal{V}_n$ can only lower its value further.

Fix d, a_0, and c, and define the oracle risk

$$L(n) = pR^c(n) + (1-p)R^\varepsilon(n).$$

Equation (4.1) implies, uniformly over integers $0 \le t \le T = \lceil c \rceil + 1$,

$$R^{\text{demo}}(n+t) - L(n) = \frac{d}{n^2}\left[p(c-\varepsilon) - t + O(n^{-1})\right].$$

Linear interpolation gives the same expansion for real $t \in [0, T]$. Let $n + \eta_n$ be the sample size at which the baseline risk equals $L(n)$. Then

$$\eta_n = p(c-\varepsilon) + O(n^{-1}).$$

When $p(c-\varepsilon) = 0$, the two risks coincide and $\eta_n = 0$ exactly. If $p = 1$, the description risk equals $L(n)$, so

$$\text{ESS}_n \longrightarrow c - \varepsilon = (1-r)c.$$

If $p < 1$, the variance decomposition gives $R^{\text{desc}}(n) = L(n) + \mathcal{V}_n \ge L(n)$. Since the baseline risk decreases with sample size, $\text{ESS}_n \le \eta_n$, proving

$$\limsup_{n\to\infty} \text{ESS}_n \le p(c-\varepsilon) = p(1-r)c.$$

Because $\mathcal{V}_n$ is nonnegative and need not vanish at the required rate, the result for unreliable descriptions is an upper bound. □

5.2 Assuming Relevance Under Misspecification

Consider the Bayesian predictor that sets $p = 1$ in its inference model and therefore returns $\mu_1^\top x$. We evaluate its risk, denoted $R^{\text{assume}}(n)$, under the data-generating model with relevance probability p. The assumed prior is misspecified when $p < 1$ and $r < 1$, because it omits the irrelevant component.

Without demonstrations, this predictor returns $m^\top x$. The moments used in the proof of Proposition 5.1 give

$$R^{\text{assume}}(0) = \mathbb{E}\|w - m\|^2 = \big(1 - (2p-1)(1-r)\big)a_0 d.$$

In each coordinate, these moments give $\mathbb{E}(w_i - m_i)^2 = a_0 - 2p(1-r)a_0 + (1-r)a_0$. For $r < 1$, this risk exceeds the baseline $a_0 d$ exactly when $p < 1/2$. At $r = 1$, the description carries no information, $m = 0$ almost surely, and the two risks coincide for every p.

With demonstrations, the predictor continues to use $\mu_1^\top x$ regardless of Z, where

$$\mu_1 = m + (cI_d + W_n)^{-1} X_n^\top (y_{1:n} - X_n m).$$

For fixed a_0 and n, $\mu_1 \to m$ in mean square as $c \to \infty$. Conditional on relevance, $\mathbb{E}\|w - m\|^2 = r a_0 d \to 0$; conditional on irrelevance, it is $(2-r)a_0 d \to 2a_0 d$. Thus

$$\lim_{c \to \infty} R^{\text{assume}}(n) = 2(1-p)a_0 d.$$

For every $p < 1$, this is a positive risk floor at fixed n, whereas $R^{\text{demo}}(n) \to 0$ as $n \to \infty$. Hence a sufficiently precise description can increase risk relative to the demonstrations-only predictor when relevance is assumed despite $p < 1$. At $p = 0.9$, $c = 100$, $d = 16$, and $a_0 = 10$, the predictor assuming relevance has risk 25.9, compared with 26.6 without the description, at $n = 14$. At $n = 15$, these risks are 25.5 and 19.5, respectively. Its risk remains higher at every larger demonstration count evaluated, up to 10^5. The order of limits matters: the preceding argument takes $c \to \infty$ at fixed n; it does not imply that demonstrations cannot overcome a fixed finite prior precision.

Chapter 6

Meta-Trained Transformers

We first test whether Transformers learn to use descriptions under the correctly specified model. Following Garg et al. (2022), we train from scratch with $d = 5$ and $a_0 = 10$. Prompts contain an optional description token carrying m, up to 20 demonstration tokens, and a query token (Huang & Ge, 2025). The description token does not state c or p; the network must learn these properties across training tasks. Half the prompts omit the description, supplying a demonstrations-only baseline for each network. We train separate networks for $c \in \{1, 10\}$ and $p \in \{1, 0.99, 0.9\}$, with two or three seeds per configuration (Section 6.1).

Matched training and test distributions. At $n = 0$, networks trained with $c = 10$, $p = 1$ achieve mean ESS 14.3 (Figure 6.1a), compared with Bayes-optimal ESS 15.9 estimated on a larger sample (Table 6.1). At lower reliability, and for $c = 1$, zero-demonstration ESS is within 0.3 of the Bayes-optimal value. Networks also learn the complementarity of partial descriptions and demonstrations. With $k = 3$ of $d = 5$ directions described at $c = 10$, ESS rises from 3.27 at $n = 0$ to 5.55 at $n = 10$, compared with Bayes-optimal values 3.22 and 5.64.

Reliability shifts. We evaluate networks at test reliabilities different from their training reliability. We compare the Bayesian predictor using the training reliability, p_{train}, with the Bayes-optimal predictor using the test reliability, p_{test} (Raventós et al., 2023). At $n = 0$, the Bayesian prediction is $p\, m^\top x$. We summarize a network's predictions by the *fitted description weight* q, estimated by regressing them on $m^\top x$. This coefficient describes prediction behavior and need not represent a probability

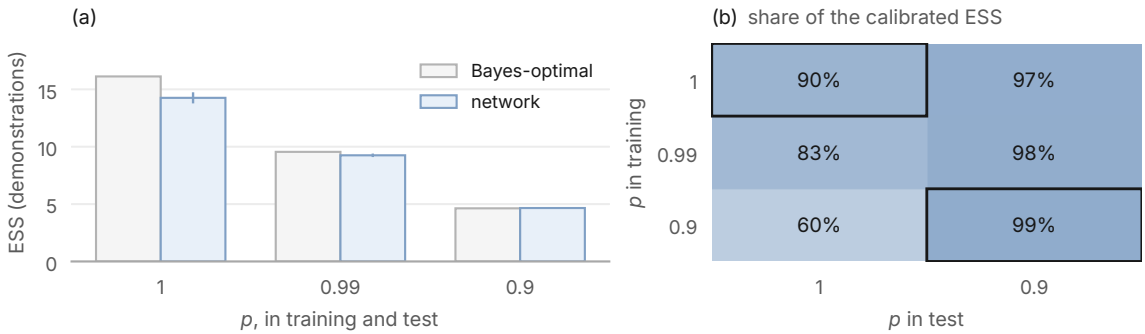


Figure 6.1: **Meta-trained Transformers approach Bayes-optimal ESS and reflect training reliability.** Results use $c = 10$, $d = 5$, $a_0 = 10$, and no demonstrations. (a) Training and test reliability match. Network ESS uses each network's own demonstrations-only curve; bars show the mean and one standard deviation across three training seeds. The Bayes-optimal comparison uses the same prompts. (b) Test reliability differs from training reliability. Cells show network ESS as a fraction of the Bayes-optimal ESS under the test distribution (labeled "calibrated"); outlined cells have matching reliabilities. Table 6.1 includes $c = 1$.

inside the network.

For $c = 10$, networks trained at $p = 0.9$ retain weights near 0.9 even when every test description is relevant (Figure 6.1b). This is consistent with the information available: m alone does not reveal the changed test reliability. When 10% of test descriptions are irrelevant, networks trained at $p = 1$ have higher error with a description than without one from six demonstrations onward, in every seed. Training at $p = 0.99$ avoids this failure across the tested range. Exposure to occasional irrelevance changes how the networks use subsequent evidence.

6.1 Additional Transformer Results and Training Details

Evaluation. We evaluate each network on 20,000 fresh prompts. The networks share these evaluation prompts, so standard deviations across training seeds exclude prompt-sampling variation. Each network is tested at its training reliability and at $p = 1$ and $p = 0.9$; the networks trained with the second and third seeds are also

tested at $p = 0.99$. At $d = 5$, we evaluate the demonstrations-only learning curve at every integer from 0 to 20, except for $c = 0.2$ and $c = 2$, which use counts from 0 to 15. We compute ESS at the first crossing of this curve with the prediction error obtained using the description, as in Definition 3.1. Table 6.1 reports results at $n = 0$, including test reliabilities that differ from the training reliability.

Comparing networks on the same prompts. We also compare each network with the Bayes-optimal predictor on the same prompts. With no demonstrations and $c = 10$, network ESS is 14.3 ± 0.5 at $p = 1$, 9.3 ± 0.2 at $p = 0.99$, and 4.66 ± 0.01 at $p = 0.9$, compared with Bayes-optimal values of 16.1, 9.6, and 4.63 on the same prompts. On 200,000 prompts, the Bayes-optimal value at $p = 1$ is 15.9, close to the limiting expression $c + d + 1 = 16$ of Proposition 4.1. At $p = 0.99$, the 200,000-prompt value, 8.96, lies below the networks' mean ESS, whereas the value on their own prompts, 9.6, lies above it. A network's ESS can exceed the Bayes-optimal ESS on the same prompts, as it does at $p = 0.9$, because it uses the network's own demonstrations-only curve rather than the Bayes-optimal one. For $c = 1$, network ESS is within 0.1 of the Bayes-optimal ESS at every reliability; at $p = 1$, for example, it is 5.85 ± 0.03, compared with 5.89.

Architecture and optimization. The Transformer has 12 layers, 8 attention heads, and width 256, matching the model size of Garg et al. (2022). We train with Adam at learning rate 10^{-4} for 40,000 steps, using batches of 1,024 newly sampled prompts and no curriculum. Following Zhu et al. (2026), the model predicts a mean and a log variance for the query label and minimizes log loss. The optimal predicted mean is the posterior mean, which also minimizes squared error. We evaluate the predicted mean using the loss in Equation (3.1). We use the optimizer of Garg et al. (2022), but unlike their model, ours uses bidirectional attention and no positional encoding. We choose $d = 5$ to keep training short.

Each demonstration token contains an input and its label. Two indicator coordinates distinguish token types, and the model uses no positional encoding. When a description is absent, its token is masked out of attention. The prediction is read from the query token. Labels and description centers are divided by $\sqrt{a_0}$, gradients are clipped to norm 1, and training uses bf16 autocast. The demonstration count is sampled uniformly from 0 to 20, except for the $c = 0.2$ and $c = 2$ experiments, which use 0 to

Table 6.1: **Transformer ESS and fitted description weights under matched and shifted reliability.** Results use $d = 5$, $a_0 = 10$, and no demonstrations, with 20,000 test prompts per network. Rows are grouped by training reliability. *Network* reports mean ESS and standard deviation across the indicated training seeds; for a single seed, the uncertainty is the standard error over prompts. The two Bayesian predictors use p_{train} and p_{test}, respectively; the latter is Bayes-optimal under the test distribution. We show one reference value when they coincide. Seed 0 was not evaluated at $p_{\text{test}} = 0.99$ for training reliabilities 1 and 0.9, so those rows average fewer seeds; their fitted weights differ from the other rows only because of the seeds included. Values using p_{test}, and coincident values using p_{train}, use 200,000 prompts, with standard errors at most 0.1. Other values using p_{train} use the networks' test prompts; sampling variation can change these estimates by up to 0.6, so a Monte Carlo estimate using p_{train} can exceed the corresponding estimate using p_{test}. *Weight* is the fitted description weight q in $\hat{y} \approx q\, m^\top x$, averaged across the available seeds. This coefficient is unrestricted and is not a relevance probability.

				ESS			
c	p_{train}	p_{test}	Seeds	Network	Bayes p_{train}	Bayes p_{test}	Weight q
1	1	1	2	5.85 ± 0.03	5.89	—	0.964
1	1	0.99	1	5.65 ± 0.06	5.66	5.61	0.991
1	1	0.9	2	4.02 ± 0.03	3.98	4.10	0.964
10	1	1	3	14.26 ± 0.48	15.91	—	1.008
10	1	0.99	2	9.38 ± 0.16	9.54	8.96	1.015
10	1	0.9	3	4.58 ± 0.03	4.54	4.70	1.008
1	0.99	1	2	5.87 ± 0.01	5.88	5.89	0.963
1	0.99	0.99	2	5.65 ± 0.01	5.61	—	0.963
1	0.99	0.9	2	4.02 ± 0.03	3.99	4.10	0.963
10	0.99	1	3	13.28 ± 0.57	16.01	15.91	0.982
10	0.99	0.99	3	9.25 ± 0.15	8.96	—	0.982
10	0.99	0.9	3	4.60 ± 0.04	4.56	4.70	0.982
1	0.9	1	2	5.67 ± 0.06	5.75	5.89	0.883
1	0.9	0.99	1	5.51 ± 0.06	5.55	5.61	0.901
1	0.9	0.9	2	4.03 ± 0.01	4.10	—	0.883
10	0.9	1	3	9.49 ± 0.44	10.79	15.91	0.876
10	0.9	0.99	2	8.08 ± 0.28	8.55	8.96	0.880
10	0.9	0.9	3	4.66 ± 0.01	4.70	—	0.876

15, and the $d = 10$ experiments, which use 0 to 40.

Layers use pre-normalization, GELU activations, feed-forward width 1,024, and no dropout. At $d = 5$, the model has 9.5 million parameters. Adam uses the PyTorch

defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$, with no weight decay and a constant learning rate. Training seeds are 0, 1, and 2 for $c = 10$, for $d = 10$, and for the partial-description experiments; 0 and 1 for $c = 1$; and 0 for $c \in \{0.2, 2\}$. A $d = 5$ network takes approximately 80 minutes to train on one NVIDIA L4 GPU, and a $d = 10$ network takes approximately 2.8 hours. Training all 25 networks takes approximately 38 GPU-hours.

Without a description, every $d = 5$ network's error is within 0.21 of the Bayes-optimal error of approximately 50 at $n = 0$, and within 0.32 on average over its training demonstration counts. At $d = 10$, the corresponding gaps are at most 0.50 relative to an error of approximately 100, and 0.41. With a description at $c = 10$, $p = 1$, and $n = 0$, network error ranges from 0.67 to 0.71, compared with 0.49 for the Bayes-optimal predictor. The networks therefore remove 98.6% of the baseline error, compared with 99.0% for the Bayes-optimal predictor.

Additional specificity levels. Four additional networks cover $c = 0.2$ and $c = 2$, each at $p = 1$ and $p = 0.9$. Their zero-demonstration ESS values are within 0.03 of the Bayes-optimal values evaluated on the same prompts: 2.64 versus 2.66, 2.16 versus 2.15, 7.32 versus 7.35, and 4.38 versus 4.37, respectively.

Higher dimension and partial descriptions. At $d = 10$, we train three seeds with description precision $c = 10$ and extend training and evaluation to 40 demonstrations so that the demonstrations-only learning curve reaches the ESS crossing. Network ESS is 17.8, compared with a Bayes-optimal value of 20.8 and the limiting expression $c + d + 1 = 21$. As at $d = 5$, the networks capture most of the description's predictive value but fall short of the Bayes-optimal ESS. The standard deviation across seeds is 0.3, and the shortfall, 3.0 demonstrations, is larger than the 1.8 at $d = 5$ against the same-prompt value of 16.1.

We also train networks with a partial description, as defined in Section 4.2. The description token contains m in the first $k = 3$ of the $d = 5$ coordinates and zeros elsewhere; the remaining task coordinates follow the baseline prior. For this $c = 10$ description, mean ESS across three seeds increases from 3.27 at $n = 0$ to 5.55 at $n = 10$. The corresponding Bayes-optimal values are 3.22 and 5.64. This increase agrees with the behavior described by Proposition 4.2. The large-sample limit of that proposition is $k(c - a_0^{-1})/d = 5.94$, and the standard deviation across seeds at $n = 0$

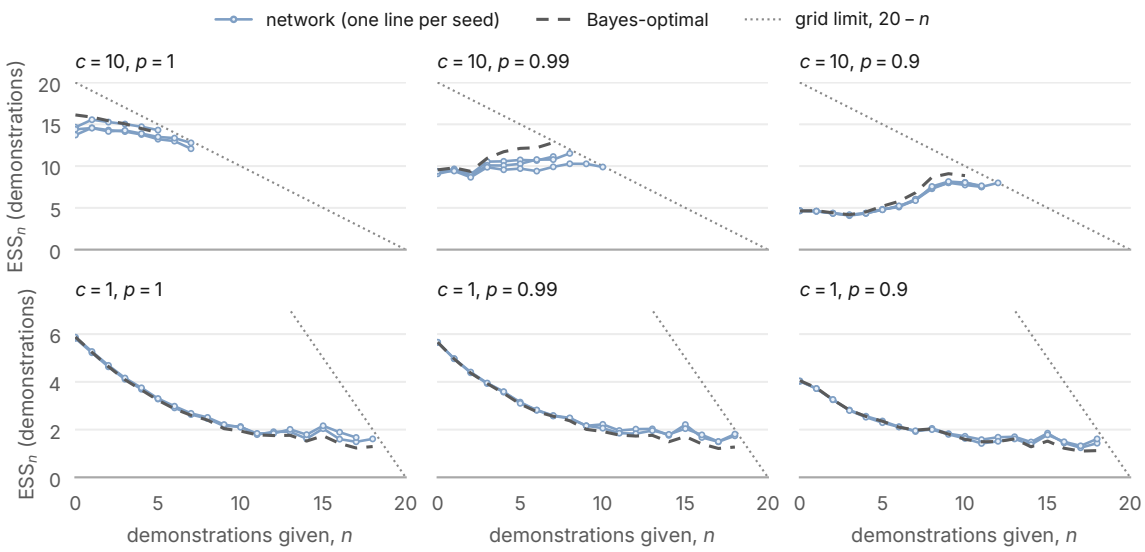


Figure 6.2: **Transformer ESS as a function of demonstration count.** Each line represents a training seed; the Bayes-optimal predictor is evaluated on the same prompts. Training and test reliability are matched. Results cover $c = 10$ and $c = 1$ at $d = 5$, using 20,000 prompts. A curve ends when its crossing exceeds the maximum of 20 demonstrations. The dotted line, $20 - n$, marks the largest ESS identifiable on this grid.

is 0.03.

ESS with demonstrations. Figure 6.2 evaluates each network at its training reliability. For $c = 1$, network ESS remains within 0.6 of the Bayes-optimal value at every demonstration count whose crossing lies within the evaluation grid. For $c = 10$, the largest shortfalls are 2.4, 2.9, and 1.4 demonstrations at $p = 1$, 0.99, and 0.9, respectively; the largest excess is 0.3. At $n = 5$, the three $c = 10$ networks trained at $p = 0.9$ have ESS values of 4.83, 4.79, and 4.77, compared with a Bayes-optimal value of 5.20 on the same prompts.

Reliability shift without demonstrations. For $c = 10$, the fitted description weights of the networks trained at $p = 0.9$ are 0.87 to 0.89 across the three seeds. Their ESS on reliable descriptions is 9.5 ± 0.4, whereas the Bayes-optimal predictor under the test distribution reaches 15.9 and the Bayesian predictor using p_{train} reaches 10.8. The networks trained at $p = 1$ or $p = 0.99$ have fitted weights between 0.95 and

1.02, so the two training reliabilities cannot be distinguished without demonstrations. At test reliability $p = 0.9$, assuming relevance costs the networks trained at $p = 1$ little at $n = 0$: their ESS is 4.58 ± 0.03, compared with 4.70 for the Bayes-optimal predictor under the test distribution.

Reliability shift after demonstrations are observed. For $c = 10$, networks trained at $p = 1$ and $p = 0.99$ both have fitted description weights near one at $n = 0$. With demonstrations, their responses to a shift in test reliability differ. At test reliability $p = 0.9$, the description reduces error for networks trained at $p = 0.99$ at every evaluated n, as it does for the Bayes-optimal predictor under the test distribution. For networks trained at $p = 1$, the description instead increases error from six demonstrations onward in every seed. The excess error ranges from 0.77 to 1.06 at $n = 6$ and peaks between 2.74 and 2.94 at $n = 9$. A Bayesian predictor that assumes relevance ($p = 1$) also first has higher error than the demonstrations-only predictor at $n = 6$, with excess error 1.12 there and 2.96 at $n = 9$.

For the networks trained at $p = 0.99$, the reduction in error is 3.5 to 3.7 with six demonstrations and 0.08 to 0.13 with 20. With eight demonstrations, for example, the first network trained at $p = 1$ has error 4.6 with the description and 2.3 without it.

Chapter 7

Pretrained Language Models

We next ask whether pretrained language models use description statistics stated in a prompt. We study numerical mean estimation, a scalar regression task with fixed input $x = 1$, followed by semantic classification. The scalar task tests the use of precision and relevance probabilities without the random-design dimension effect.

7.1 Task and Evaluation

Each numerical task has a hidden mean $w \sim \mathcal{N}(500, 100^2)$ and demonstrations drawn from $\mathcal{N}(w, \sigma^2)$, with $\sigma = 60$, then rounded to integers. A description gives a center m and standard deviation τ: "Its mean was drawn from a normal distribution with mean 510 and standard deviation 30." Before rounding and range restrictions, the relevant component satisfies $w \mid m \sim \mathcal{N}(m, \tau^2)$. We measure description precision relative to one observation, $c = \sigma^2/\tau^2$, retaining the unit-noise convention of the theory. Stated standard deviations 60, 30, and 10 correspond to $c = 1$, 4, and 36.

We vary stated precision and add warnings of a 50% chance that the description is "wrong." These warnings change the stated reliability while the descriptions remain relevant. In a separate fixed-value control, we hold m and all demonstrations fixed and describe a center generated at $c = 36$ as having $c = 4$. This isolates the response to stated precision. Section 7.9 also evaluates irrelevant descriptions and actual mixtures of the two relevance branches.

The main comparison uses the same 200 tasks for five models with reasoning enabled and disabled: gpt-oss-20b (OpenAI, 2025), Qwen3.5 9B (Qwen Team, 2026),

Gemma 4 31B (Gemma Team, 2026), GPT-6 Sol (OpenAI, 2026), and Claude Opus 4.6 (Anthropic, 2026b). Claude Haiku 5.5 (Anthropic, 2026a) is evaluated with reasoning at medium effort. "Reasoning enabled" denotes the configured reasoning mode; individual replies may still contain no recorded reasoning output. Paired runs use the same model identifier, although serving and decoding settings differ for some models. Sections 7.5–7.10 give execution and readout details and a larger run without reasoning on 2,000 tasks.

Bayesian reference. Our *prompt-based Bayesian reference* uses the observation noise and description uncertainty stated in the prompt. When the mean is described as unknown, it assumes a uniform prior over the integers 100 to 999, both without a description and under irrelevance. When the prompt asserts relevance, its prediction is approximately the precision-weighted Gaussian posterior mean (Neuenschwander et al., 2020),

$$\hat{w}_{\text{ref}} = \frac{c\,m + n\bar{y}}{c + n}, \tag{7.1}$$

where $\bar{y}$ is the demonstration mean. Rounding and boundary effects are negligible for this expression. In the idealized continuous Gaussian model with a flat baseline prior, a correctly specified description has exact ESS c. Every observation measures the same scalar mean, so there is no random-design dimension contribution.

Intuitively, the description counts as c further demonstrations at m: it adds a precision of $1/\tau^2 = c/\sigma^2$, equal to that of c demonstrations. This is the conjugate normal model of Chapter 2 with $n_0 = c$, and the squared error is $\sigma^2/(c+n)$ with the description and σ^2/n without it.

The reference follows the stated precision and reliability even when they differ from the generating values. Agreement with it measures how models use the prompt's statistical information; it need not imply optimal prediction under the evaluation distribution. When relevance is uncertain, the posterior probabilities also depend on the prior under irrelevance, which the original prompt leaves unspecified (Section 7.11).

We compute ESS using each model's demonstrations-only curve and report the median over $n \in \{1, 2, 4\}$. Applied to the reference, this estimator gives 34.6 at $c = 36$, rather than the idealized 36, because of finite-task sampling and interpolation. We also regress predictions on m, $\bar{y}$, and an intercept. The coefficient on m is the *fitted description weight*; the reference weight is $c/(c+n)$ when relevance is asserted.

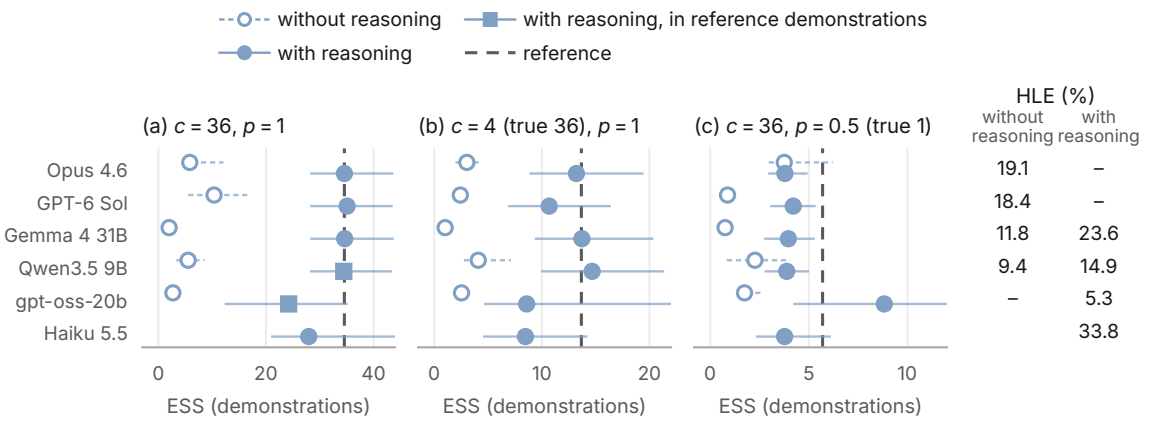


Figure 7.1: **Reasoning improves the use of stated precision.** Summary ESS on 200 tasks at $\sigma = 60$ (median over $n = 1, 2, 4$), with paired 95% task-bootstrap intervals. Open points indicate reasoning disabled; filled points indicate reasoning enabled; vertical dashed lines show the prompt-based Bayesian reference. (a) Relevant descriptions generated and stated at $c = 36$. (b) The same centers stated as $c = 4$: reference ESS is 13.7, because the centers were generated at $c = 36$. (c) The same relevant descriptions with stated reliability $p = 0.5$: reference ESS is 5.7 under the assumed uniform prior for the irrelevant component. For GPT-6 Sol and Haiku 5.5, stating that prior in the prompt moves their predictions from averaging with prior weights toward the reference and raises their ESS by at most 1.2 (Section 7.11). Squares in (a) use the reference's demonstrations-only curve instead of each model's curve and report a different comparison. Intervals are clipped to the display; three upper endpoints lie beyond the grid. HLE scores (Center for AI Safety et al., 2026; Artificial Analysis, 2026) provide contextual information; dashes denote unavailable scores and blanks denote modes not evaluated.

7.2 Reasoning and Stated Precision

Descriptions are underweighted without reasoning. For $c = 36$, summary ESS ranges from 2.0 to 10.3 across the five models without reasoning, compared with 34.6 for the reference (Figure 7.1a). Fitted weights likewise show substantially less reliance on the description. Sensitivity to stated precision varies: describing the same center as $c = 4$ instead of $c = 36$ reduces reference ESS from 34.6 to 13.7, GPT-6 Sol's from 10.3 to 2.4, and Opus 4.6's and Gemma 4 31B's by about half; Qwen3.5 9B and gpt-oss-20b change less. At $n = 4$, GPT-6 Sol's fitted weights are 0.47 and 0.19 at stated $c = 36$ and 4, and Opus 4.6's are 0.40 and 0.20, against reference weights 0.90 and 0.50.

Reasoning improves the use of stated precision. With reasoning enabled, Opus 4.6 and Gemma 4 31B attain approximately the reference ESS in both conditions, and GPT-6 Sol comes close. Haiku 5.5 at medium effort falls short (28 and 8.5, against 34.6 and 13.7), mainly at $n = 1$, where 21–28% of replies contain no recorded reasoning output. On tasks where m and $\bar{y}$ are sufficiently separated, 89–100% of these four models' replies at $n = 4$ imply description weights within 0.05 of the reference (Section 7.11).

Qwen3.5 9B and gpt-oss-20b also approach the reference weights, despite lower reported HLE scores (Section 7.10). Their demonstrations-only curves flatten at larger n, and gpt-oss-20b makes substantial averaging errors. Some ESS crossings therefore fall beyond the grid. Squares in Figure 7.1 use the reference's demonstrations-only curve and report $t^*_{\text{ref}} - n$, where that curve reaches the model's error at t^*_{ref}. The remaining points use each model's own baseline.

7.3 Stated Reliability

A stated 50% chance of irrelevance reduces reference ESS from 34.6 to 5.7 under the uniform baseline prior. Without reasoning, most models show a much smaller reduction; GPT-6 Sol falls from 10.3 to 0.9. With reasoning, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B closely match predictions obtained by *averaging with prior weights*. Each component prediction conditions on the demonstrations, while its weight remains p or $1-p$. In the scalar approximation, the resulting weight on m is $pc/(c+n)$

(Table 7.7).

Bayesian model averaging uses posterior component probabilities, which depend on both component marginal likelihoods. The original warning leaves the baseline prior unspecified, saying only that the mean is otherwise "unknown." We therefore retest GPT-6 Sol at low effort and Haiku 5.5 at medium effort with that prior stated explicitly: the mean "was instead drawn uniformly at random from the whole numbers 100 to 999."

With the baseline prior stated and a 50% probability of irrelevance, both models' predictions are closer to the Bayesian reference. On tasks where posterior and prior weighting give predictions that differ by at least 5, the share of replies closer to the reference rises from 0–5% to 92–97% for GPT-6 Sol and from 21–41% to 87–95% for Haiku 5.5.

To test whether description weights vary with the evidence, we divide each reply's description weight by $c/(c+n)$. Under the scalar approximation in Equation (7.1), the reference's normalized weight equals its posterior relevance probability, π_n. Regressing the models' normalized weights on π_n gives slopes 0.96 and 0.98 for GPT-6 Sol and Haiku 5.5, respectively, compared with 0.01 and 0.22 under the original warning (Section 7.11). In separate wording experiments, both models often repeat m under "give or take" and "±" formulations, which do not define a prior standard deviation.

7.4 A Semantic Task

We next consider a message-filing task with a uniform prior over 60 hidden policies. Policy notes express constraints in ordinary language: a broad note leaves 20 compatible policies, and a specific note leaves 2. The model must interpret the note to determine which policies remain possible (Section 7.13).

With reasoning, Haiku 5.5 and GPT-6 Sol obtain ESS 9.9 and 9.5 for the specific note at $n = 2$, compared with 11.4 for the reference. For unreliable notes, the generator draws the note independently of the task with probability 0.5, and the prompt specifies this component. We compare Bayesian predictions with those obtained by retaining the prior component weights, except when the demonstrations rule out the note. Among queries where these predictions differ detectably, 77–99% of the two models' predictions are closer to the Bayesian reference. The agreement with posterior weighting therefore extends to ordinary-language constraints within this finite task family.

7.5 Additional Language-Model Results and Evaluation Details

We generate 2,000 tasks. Six open-weight models answer all tasks without reasoning; we refer to these experiments as the *full run*. The models are gpt-oss-20b, Qwen3.5 9B, and Gemma 4 31B, which also appear in Sections 7.1–7.3, together with Phi-4 (14B; Abdin et al., 2024), OLMo 2 32B Instruct (Team OLMo et al., 2025), and Gemma 4 12B. The main numerical experiments in Sections 7.1–7.3 use the first 200 of these tasks. The semantic experiment uses a separate set of 200 tasks, described in Section 7.13. Unless stated otherwise, the results in this section and those that follow use the full run without reasoning.

Table 7.1 reports summary ESS, defined as the median over $n \in \{1, 2, 4\}$, with uncertainty intervals. Figure 7.2 shows the underlying learning curves. Tables A.1 and A.2, in Appendix A, report ESS at every evaluated demonstration count; Table 7.2 compares interpolation methods; Table A.3, also in Appendix A, reports description weights for the models with a distribution readout; and Table 7.3 evaluates mixtures of relevant and irrelevant descriptions.

For the numerical experiments, the *prompt-based Bayesian reference* combines the probabilistic statements in the prompt with an assumed baseline prior when the prompt does not specify one. It is optimal under that assumed model, but need not minimize error under the actual task distribution when the two differ. We therefore distinguish this reference from the correctly specified Bayesian predictor in the theory. The semantic experiment specifies a finite task distribution and uses its exact Bayesian predictor.

In the numerical task, $c = \sigma^2/\tau^2$ is the prior precision expressed in units of one demonstration. We distinguish the precision used to generate a description value from the precision stated in the prompt whenever they differ. Similarly, a stated reliability supplies a prior relevance probability to the model; it need not equal the proportion of relevant descriptions in the evaluation sample.

7.6 Prompts, Tasks, and Model Readouts

Model inclusion. We retain a model only if it produces a number in at least three quarters of replies overall and, averaged over $n = 1$, 2, and 4 and both noise levels, in at least half of replies without a description. The latter responses determine the baseline curve used for summary ESS. This criterion excludes MiniCPM5 2B (MiniCPM Team, 2026), OLMo 2 7B Instruct, and OLMo 2 13B Instruct, whose replies begin with a number in 62%, 20%, and 10% of cases, respectively. These models often begin with explanations such as "To predict…" or "Given…" despite the numeric-only instruction. For example, OLMo 2 7B's summary ESS for the $c = 36$ description is 21, but this value comes from a readout that rests on a minority of the model's probability.

The full run uses Hugging Face Transformers at each model's native precision, with checkpoints `openai/gpt-oss-20b`, `microsoft/phi-4`, `allenai/OLMo-2-0325-32B-Instruct`, `Qwen/Qwen3.5-9B`, `google/gemma-4-12B-it`, and `google/gemma-4-31B-it`. Tasks are sampled by a seeded generator. Reasoning is disabled where the chat template supports this option. For gpt-oss-20b, generation begins in the final channel, after the channel normally used for reasoning.

Distribution and written readouts. For gpt-oss-20b, Phi-4, and OLMo 2 32B without reasoning, the prediction is the mean of the first-token distribution, renormalized over the 900 three-digit number tokens. We call this the *distribution readout*. Other models, and all reasoning runs, use the number in the final answer, called the *written readout*. We also evaluate written readouts for the three distribution-readout models as a robustness check (Table 7.1 for gpt-oss-20b, and Table 7.4). OLMo 2 32B assigns 41% of first-token probability to numbers without a description at $n = 1$, and 75% at $n = 2$. Its distribution readout at $n = 1$ therefore summarizes a minority of the first-token probability mass.

Prompt. The following text is supplied as the user turn in the model's chat template:

```
The numbers below are drawn from a normal distribution with standard deviation 60.
Its mean was drawn from a normal distribution with mean 510 and standard deviation 30.

548
431

Predict the next number.  Reply with the number only.
```

Table 7.1: **Description ESS and weights in the full run without reasoning.** Left: median ESS over $n = 1$, 2, and 4, for relevant descriptions with $c = 1$, 4, and 36, and for the $c = 36$ description when irrelevant or accompanied by a stated 10% or 50% chance of being wrong. Brackets give 95% bootstrap intervals. The censoring marker $\leq$ indicates that one demonstration alone already outperforms the description-conditioned prediction at a boundary used in the summary. Right: the coefficient on the stated value at $n = 1$; the Bayesian reference weight is $c/(c + n)$. All results use $\sigma = 60$. "Written" denotes the written readout.

	ESS, median over $n = 1$, 2, 4						weight, $n = 1$		
	relevant			$c = 36$			relevant		
Model	$c = 1$	$c = 4$	$c = 36$	irrelevant	10% wrong	50% wrong	$c = 1$	$c = 4$	$c = 36$
Reference	1.3	4.4	36	≤ -1.0	22	5.8	0.50	0.80	0.97
gpt-oss-20b	0.9 [0.7, 1.0]	2.0 [1.7, 2.3]	2.8 [2.6, 3.1]	≤ -0.4	2.4 [1.9, 2.6]	2.3 [1.9, 2.7]	0.53	0.48	0.49
Phi-4	0.7 [0.4, 0.9]	4.1 [3.5, 4.6]	7.6 [6.5, 8.6]	≤ -1.0	8.7 [7.8, 9.8]	9.7 [8.7, 11]	0.82	0.76	0.74
OLMo 2 32B	1.9 [1.7, 2.1]	3.4 [3.2, 3.9]	5.3 [4.5, 6.3]	≤ -0.6	6.0 [5.6, 7.1]	5.4 [5.0, 6.0]	0.72	0.70	0.70
gpt-oss-20b (written)	0.2 [0.1, 0.4]	2.0 [1.9, 2.6]	3.7 [3.0, 4.4]	≤ -1.0	2.4 [2.1, 2.6]	2.3 [2.1, 2.6]	0.64	0.52	0.52
Qwen3.5 9B (written)	0.2 [−0.1, 0.4]	2.7 [2.1, 3.2]	4.0 [3.5, 4.5]	≤ -1.0	0.6 [0.3, 1.0]	1.0 [0.6, 1.5]	0.81	0.76	0.80
Gemma 4 12B (written)	0.0 [−0.1, 0.1]	0.5 [0.3, 0.7]	1.8 [1.4, 2.2]	≤ -0.7	1.3 [1.1, 1.6]	2.7 [2.5, 3.1]	0.34	0.24	0.59
Gemma 4 31B (written)	0.0 [0.0, 0.0]	0.2 [0.1, 0.3]	1.4 [1.2, 1.6]	≤ -0.7	0.6 [0.5, 0.7]	0.5 [0.4, 0.6]	0.48	0.49	0.72

Without a description, the second sentence is "Its mean is unknown." To state unreliability, a third sentence is added: "There is a 10% chance that this statement is wrong, in which case the mean is unknown." In the rerun that states the baseline prior, the final clause becomes "in which case the mean was instead drawn uniformly at random from the whole numbers 100 to 999." With no demonstrations, the final question is "Predict the first number."

Task generation. A relevant description value is sampled conditional on the task so that the joint distribution satisfies $m \sim \mathcal{N}(500, 100^2 - \tau^2)$ and $w \mid m \sim \mathcal{N}(m, \tau^2)$. A task is redrawn if a displayed number or description value falls outside 100 to 999; this occurs for less than 1% of tasks. The full run also includes $\sigma = 30$ conditions. Both standard deviations stated in the prompt are then halved, preserving $c \in \{1, 4, 36\}$. The qualitative pattern is unchanged (Table A.1).

Numeric response rates. For the distribution readout, the numeric response rate is the probability mass assigned to number tokens at the first position. For the written readout, it is the fraction of replies containing a three-digit number. Tables 7.4, A.1,

A.2, and A.3 report this quantity as "number". Tables 7.4 and A.3 italicize cells whose rate is below three quarters.

Written-readout implementation and checks. The written readout extracts the number in the last `\boxed{}` expression in the final answer, if present. Otherwise, it takes the last number with three digits before an optional decimal part, retaining the decimals. A digit group after a thousands separator, as in 14,945, does not count as a qualifying number. In the full run, decoding is greedy with a limit of eight new tokens, so the reply is the number itself. Those predictions were recorded using an earlier readout that took the first qualifying number and retained only its integer part. Replies without a qualifying number are excluded from both error curves and weight regressions; the reported response rates identify the extent of this exclusion.

For gpt-oss-20b, written-readout weights differ from distribution-readout weights by at most 0.21 across all conditions and demonstration counts. For $c = 1$, 4, and 36, written-readout summary ESS is 0.2, 2.0, and 3.7, compared with 0.9, 2.0, and 2.8 from the distribution readout. A single greedy response is a noisier summary than the distribution mean. Phi-4 produces a number in only 58% of written replies across all conditions, so Table 7.1 reports only its distribution readout. In the wording experiment, however, its numeric response rate exceeds three quarters for the original wording. OLMo 2 32B produces a number in only 5%, 36%, and 48% of written replies without a description at $n = 1$, 2, and 4, respectively, with $\sigma = 60$; we therefore do not report its written-readout ESS. Its written-readout weights for the three relevant descriptions at $n = 1$ are 0.83, 0.81, and 0.83, compared with distribution-readout weights of 0.72, 0.70, and 0.70.

7.7 ESS Estimation and the Bayesian Reference

Measurement. We apply Definition 3.1 to squared error against the hidden mean. In the full run, the baseline curve is evaluated at $n = 1, 2, 4, \ldots, 64$. We interpolate squared error, rather than its square root, linearly in $\log n$. The ESS at n is the first interpolated demonstration count t that matches the description-conditioned error, minus n.

If the description-conditioned error lies below the entire baseline curve up to $t_{\max}$, we report $\mathrm{ESS}_n > t_{\max} - n$. Because the curve starts at one demonstration, we use

the marker $\mathrm{ESS}_n \leq 1 - n$ when one demonstration alone already gives lower error. These markers record censoring at the evaluation boundaries; they do not define an extrapolated matching count or an exact ESS when the crossing does not exist. Summary ESS uses $n = 1$, 2, and 4. We exclude $n = 0$ because models rarely return a number without a description or demonstrations. At large n, ESS is strongly affected by slow baseline learning and excessive weight on the $c = 1$ and $c = 4$ descriptions. At $\sigma = 60$ and $n = 64$, model root-mean-square error without a description ranges from 9 to 20, compared with 7.5 for the reference. A learner with a flatter baseline curve can assign a larger ESS to the same error reduction. Description-weight estimates complement ESS because they do not use that curve.

Bootstrap replicates resample tasks jointly for the description-conditioned and baseline curves and recompute the complete ESS statistic. The intervals therefore measure variation across tasks; they do not include interpolation bias, quantified below. We use 500 replicates with a fixed seed and the same resampled tasks for every model.

Reference prior and exact sample equivalence. The prompt-based Bayesian reference takes "unknown" to mean uniform over 100 to 999. Without a description, its prediction is therefore approximately the demonstration mean, with negligible boundary truncation for the evaluated prompts. For a relevant Gaussian description with correctly stated precision, the idealized continuous model with a flat baseline prior has exact sample equivalence c. Integer rounding, finite-range truncation, and the task-rejection rule are not part of that exact identity; we compute the reported reference values on the same generated tasks as the models. When a chance of being wrong is stated, the reference uses it as the prior probability of irrelevance and computes the posterior relevance probability from the demonstrations. Otherwise, it treats the description as relevant.

Applying the empirical ESS procedure to the full-run reference gives 1.3, 4.4, and 36 for $c = 1$, 4, and 36 at $\sigma = 60$, and 1.2, 4.7, and 43 at $\sigma = 30$. Interpolation in $\log n$ induces an upward bias, while finite-task sampling can shift the estimate in either direction. We use the same estimator for the reference and the models. On the 200 tasks of Sections 7.1–7.3, the grid additionally includes $n = 40$ and 48 near the $c = 36$ crossing, and the reference values are 1.0, 3.5, and 34.6. With a stated chance of irrelevance, the reference's summary ESS varies more across task samples: on 20,000

tasks it is 7.3 at a stated 50% and 23.4 at 10%, compared with 5.8 and 22 on the full run's 2,000 tasks and 5.7 and 27.5 on the first 200. Each model is compared with the reference on the same tasks.

The actual task prior, $\mathcal{N}(500, 100^2)$, is not stated in the prompt. A reference using that prior would have baseline precision equivalent to 0.36 demonstrations at $\sigma = 60$ and 0.09 at $\sigma = 30$. At $\sigma = 60$, the corresponding idealized continuous Gaussian model would give a relevant description exact ESS $c - 0.36$, namely 0.64, 3.64, and 35.64. This is the scalar analogue of $c - a_0^{-1}$ in Chapter 4. We use the prompt-based reference because the models are not given the actual task prior. This distinction matters under misspecification: unlike the correctly specified Bayesian predictor in Chapters 4–5, a reference that incorrectly assumes an irrelevant description to be relevant can have negative ESS, indicating increased prediction error (Reimherr et al., 2021).

Sensitivity to interpolation. The full-run grid doubles n, so interpolation can affect ESS. On the reference's idealized errors, σ^2/n without a description and $\sigma^2/(c+n)$ with one, interpolation in $\log n$ overestimates ESS by 0.2, 0.3, and 1.8 for $c = 1$, 4, and 36. Linear interpolation in n gives overestimates of 0.3, 0.6, and 4.1.

For the models without reasoning, the relevant crossings are unique. Among the twelve baseline curves across the two noise levels, only Phi-4 at $\sigma = 60$ increases, slightly, from 20.3 at $n = 32$ to 20.4 at $n = 64$. Table 7.2 compares four estimators: interpolation in $\log n$; linear interpolation in n; a least-squares non-increasing fit using the pool-adjacent-violators algorithm; and a relative-least-squares fit of $A/(n+n_0)+B$ to the baseline errors. The final form is a Bayesian learning curve plus an error floor and recovers exact ESS when fit to the reference's idealized errors.

Across these choices, no model's summary ESS changes by more than 1.4 demonstrations. For the $c = 36$ description, no model exceeds 8.1, while the reference ranges from 35 to 38. Differences become larger at higher demonstration counts, where a flat learning curve turns a small error difference into a large change in ESS. For example, gpt-oss-20b at $n = 8$ and $c = 36$ has ESS 17 under the estimator in Table A.1, but 9.2 under the fitted curve.

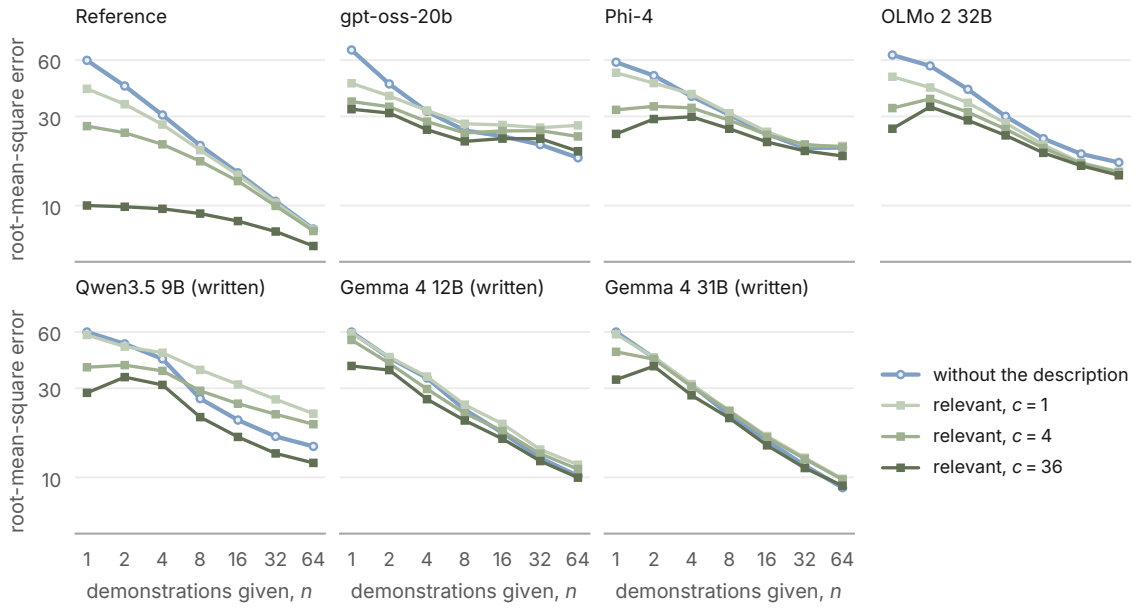


Figure 7.2: **Learning curves used to estimate language-model ESS.** Root-mean-square error versus demonstration count for the reference and six models in the full run without reasoning, at $\sigma = 60$. Blue curves omit the description; green curves use relevant descriptions, with darker shades indicating higher c. Replies without a number are excluded. ESS is the horizontal distance from the description-conditioned point at n to the baseline curve at the same error. For four models, error with $c = 36$ increases between one and two demonstrations because the models shift toward the observations more than the reference does (Table A.3).

Table 7.2: **Sensitivity of summary ESS to baseline-curve estimation.** Median ESS over $n = 1$, 2, and 4 for relevant descriptions, using the full run without reasoning at $\sigma = 60$. "Log n" is the primary estimator; "linear" interpolates in n; "monotone" applies a non-increasing fit; and "fit" uses $A/(n + n_0) + B$. The reference's idealized ESS values are 1, 4, and 36.

	$c = 1$				$c = 4$				$c = 36$			
Predictor	log n	linear	monotone	fit	log n	linear	monotone	fit	log n	linear	monotone	fit
Reference	1.3	1.4	1.3	1.0	4.4	4.7	4.4	4.0	36	38	36	35
gpt-oss-20b	0.9	1.0	0.9	0.9	2.0	2.4	2.0	2.4	2.8	2.9	2.8	3.4
Phi-4	0.7	0.8	0.7	0.4	4.1	4.4	4.1	3.6	7.6	8.1	7.6	7.1
OLMo 2 32B	1.9	2.1	1.9	1.4	3.4	3.7	3.4	3.3	5.3	5.8	5.3	5.5
Qwen3.5 9B (written)	0.2	0.3	0.2	−0.1	2.7	2.9	2.7	1.3	4.0	4.3	4.0	2.8
Gemma 4 12B (written)	0.0	0.0	0.0	0.0	0.5	0.6	0.5	0.4	1.8	1.9	1.8	1.6
Gemma 4 31B (written)	0.0	0.0	0.0	−0.1	0.2	0.3	0.2	0.0	1.4	1.7	1.4	1.0

7.8 Results of the Full Run

Summary ESS. Across the six models, the $c = 36$ description has summary ESS 1.4 to 7.6, compared with reference ESS 36 on these tasks, and ESS at most 15 at any of $n = 1$, 2, and 4 (Table 7.1). The $c = 1$ and $c = 4$ descriptions have summary ESS 0.0 to 1.9 and 0.2 to 4.1, compared with 1.3 and 4.4. ESS rises with c mainly because a more precise description has a value closer to the task, as Table 7.5 shows.

Description weights. At $\sigma = 60$, with no demonstrations and no stated reliability, the models return the description value (weight 0.92 to 1.10), although Phi-4 and OLMo 2 32B then begin only 20 to 23% and 50 to 59% of their replies with a number. At $n = 1$, the weight of gpt-oss-20b, Phi-4, and OLMo 2 32B changes by at most 0.08 from $c = 1$ to $c = 36$, and if anything falls, whereas the reference weight rises from 0.50 to 0.80 to 0.97 (Table 7.1). At every n from 1 to 64 and at both noise levels, the spread of these three models' weights across c is at most 0.16, compared with a reference spread of 0.34 to 0.71 (Table A.3). The weights of the two Gemma models do rise with c, by about 0.25 between $c = 1$ and $c = 36$ at $n = 1$, half the reference rise. The weight of Qwen3.5 9B does not. Models therefore underweight a precise description and, with many demonstrations, overweight the least precise one. At $n = 64$, gpt-oss-20b still assigns weight 0.23 to the $c = 1$ value, compared with reference weight 0.02, and that description raises its error (ESS -57).

Stated unreliability. A stated 10% or 50% chance of being wrong leaves the ESS of gpt-oss-20b and OLMo 2 32B about where it was, and Phi-4's ESS even rises, from 7.6 to 8.7 and 9.7. Qwen3.5 9B and Gemma 4 31B have lower ESS, but by about as much for 50% as for 10%. Gemma 4 12B's ESS falls at 10% and rises at 50% (Table 7.1). No model's ESS falls in proportion to the stated probability, whereas reference ESS falls to 22 and 5.8.

7.9 Unreliability, Generating Precision, and Wording

Irrelevant descriptions. An irrelevant description uses an independently drawn value. At $c = 36$, these descriptions have negative ESS for every model and for the prompt-based reference, which treats them as relevant. The models assign less weight

to the description and consequently incur lower error in this condition: at $n = 4$, their root-mean-square errors range from 35 to 64, compared with 124 for the reference. Also at $n = 4$, their weight on an irrelevant $c = 36$ value is 0.01 to 0.33, compared with 0.18 to 0.44 on a relevant one.

Actual unreliability. The stated-unreliability experiments keep descriptions relevant and vary what models are told. Table 7.3 instead evaluates a mixture of relevant and irrelevant descriptions, following the construction in Chapter 3. We combine squared errors from the relevant and irrelevant conditions, on the same tasks, with weights p and $1 - p$, and then estimate ESS. We evaluate models both with and without disclosure of the irrelevance probability. We compare a reference that sets $p = 1$ with one that uses the actual reliability and updates its posterior relevance probability from the demonstrations. Both retain the numerical reference's uniform baseline prior, rather than the Gaussian prior of the task generator. Thus, matching the reliability does not make the reference correctly specified for the task generator.

At actual reliability $p = 0.9$, a $c = 36$ description has ESS 5.5 for the reference that uses the actual reliability and 0.2 for the reference assuming relevance. Model ESS ranges from 0.4 to 3.2 when the reliability is not stated and from -0.1 to 4.0 when it is stated. Model weights depend weakly on p, except for Gemma 4 12B at $n = 1$, and stating the probability changes ESS by at most two demonstrations in either direction. Their nearly fixed weight on the description thus lies between assuming relevance and using the actual reliability. At $p = 0.5$, the description is worth less than one demonstration to the reference that uses the actual reliability and at most one to every model.

Wording without reasoning. We repeat the three relevant-description conditions and the no-description baseline at $n = 1$, 2, and 4 using three alternative description sentences: "Its mean is about m, give or take τ.", "Mean: $m \pm \tau$", and "An earlier estimate put its mean at m, with a standard error of τ." Table 7.4 uses the full-run tasks and written readouts.

Wording substantially changes the weight assigned to the description. For Gemma 4 12B with $c = 1$ and $n = 1$, this weight is 0.34 under the original wording and 1.00 under each alternative. For gpt-oss-20b, it ranges from 0.17 to 0.67. A fitted coefficient above 1 means that, holding the demonstration mean fixed in the regression, a unit change

Table 7.3: **Description ESS under actual unreliability.** Relevant and irrelevant conditions at $c = 4$ and $c = 36$ are combined with weights p and $1 - p$. Results are summary ESS with 95% task-bootstrap intervals for the full run without reasoning at $\sigma = 60$. "Assume relevance" fixes $p = 1$; "Use actual p" uses the generating reliability and updates relevance from demonstrations. Both retain the reference's uniform prior for the irrelevant component. A bound indicates that one demonstration alone outperforms the description-conditioned prediction at one or more summary demonstration counts.

	$p = 0.9$ (10% irrelevant)		$p = 0.5$ (50% irrelevant)	
Predictor	$c = 4$	$c = 36$	$c = 4$	$c = 36$
Assume relevance	1.0 [0.8, 1.1]	0.2 [0.0, 0.4]	≤ -1.0	≤ -1.0
Use actual p	2.3 [2.1, 2.7]	5.5 [5.1, 6.0]	0.6 [0.5, 0.7]	0.9 [0.8, 1.1]
gpt-oss-20b, not told	1.2 [1.0, 1.4]	1.6 [1.4, 1.7]	0.1 [0.0, 0.2]	≤ 0.0
gpt-oss-20b, told	1.1 [1.0, 1.3]	1.3 [1.1, 1.4]	-0.1 [-0.1, 0.0]	-0.1 [-0.1, 0.0]
Phi-4, not told	1.3 [1.1, 1.5]	1.9 [1.7, 2.4]	≤ -1.0	≤ -1.0
Phi-4, told	2.7 [2.2, 3.2]	3.6 [3.1, 4.1]	≤ 0.0	≤ 0.0
OLMo 2 32B, not told	2.5 [2.2, 2.8]	3.2 [2.9, 3.5]	≤ 0.7	≤ 0.6
OLMo 2 32B, told	3.2 [2.9, 3.4]	4.0 [3.6, 4.3]	≤ 1.0	≤ 0.9
Qwen3.5 9B (written), not told	1.0 [0.7, 1.2]	1.8 [1.3, 2.1]	≤ -0.3	≤ -0.9
Qwen3.5 9B (written), told	-1.0 [-1.0, -0.7]	-0.1 [-0.5, 0.3]	≤ -1.0	≤ -1.0
Gemma 4 12B (written), not told	0.3 [0.2, 0.5]	0.7 [0.5, 1.0]	≤ -0.1	≤ 0.0
Gemma 4 12B (written), told	0.6 [0.5, 0.7]	1.0 [0.9, 1.3]	≤ 0.0	≤ 0.1
Gemma 4 31B (written), not told	0.1 [0.0, 0.2]	0.4 [0.3, 0.5]	≤ 0.0	≤ -0.1
Gemma 4 31B (written), told	≤ 0.0	0.2 [0.1, 0.3]	≤ 0.0	≤ 0.0

in the description value corresponds to more than a unit change in the prediction. Nevertheless, none of the tested wordings yields the reference's sensitivity to prior precision: across models and wordings, the weight difference between $c = 36$ and $c = 1$ is at most 0.25 at $n = 1$ and 0.44 at $n = 4$, compared with reference differences of 0.47 and 0.70. The informal descriptions do not uniquely specify a probability distribution, so this comparison also varies how precisely the task is specified.

Table 7.4: **Description weights under four wordings without reasoning.** Weights are reported for the three relevant descriptions at $n = 1$ and 4, using written readouts on the full-run tasks. "Number" denotes the numeric response rate averaged over demonstration counts; a weight is italicized when the response rate at that count is below three quarters. The reference weights are 0.50, 0.80, and 0.97 at $n = 1$, and 0.20, 0.50, and 0.90 at $n = 4$. Wording changes overall reliance on the description, but the observed change with stated precision remains smaller than the reference change.

			weight, $n = 1$			weight, $n = 4$		
Model	Wording of the description	number	$c = 1$	$c = 4$	$c = 36$	$c = 1$	$c = 4$	$c = 36$
Gemma 4 12B	normal, mean m, sd τ (ours)	1.00	0.34	0.24	0.59	0.19	0.24	0.31
	about m, give or take τ	1.00	1.00	1.00	1.00	0.87	0.76	0.82
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.60	0.66	0.74
	earlier estimate m, s.e. τ	1.00	1.00	1.00	1.00	0.45	0.54	0.67
Gemma 4 31B	normal, mean m, sd τ (ours)	1.00	0.48	0.49	0.72	0.02	0.04	0.18
	about m, give or take τ	1.00	1.00	1.00	1.00	1.01	1.22	1.45
	Mean: $m \pm \tau$	1.00	1.00	1.00	1.00	0.90	1.03	1.15
	earlier estimate m, s.e. τ	1.00	0.94	0.94	0.96	0.09	0.23	0.41
gpt-oss-20b	normal, mean m, sd τ (ours)	1.00	0.64	0.52	0.52	0.23	0.29	0.34
	about m, give or take τ	1.00	0.24	0.17	0.38	0.28	0.29	0.38
	Mean: $m \pm \tau$	1.00	0.17	0.26	0.40	0.23	0.24	0.30
	earlier estimate m, s.e. τ	1.00	0.67	0.63	0.67	0.28	0.37	0.42
OLMo 2 32B	normal, mean m, sd τ (ours)	0.61	0.83	0.81	0.83	*0.20*	*0.29*	0.38
	about m, give or take τ	0.82	0.93	0.92	0.96	0.60	0.57	0.65
	Mean: $m \pm \tau$	0.82	0.88	0.83	0.88	0.44	0.46	0.55
	earlier estimate m, s.e. τ	0.82	0.95	0.94	0.95	0.49	0.57	0.65
Phi-4	normal, mean m, sd τ (ours)	0.77	0.96	0.78	0.76	*0.38*	*0.37*	*0.38*
	about m, give or take τ	0.97	1.02	1.01	1.02	0.65	0.57	0.62
	Mean: $m \pm \tau$	0.83	*0.99*	*0.87*	*0.87*	0.45	0.42	0.50
	earlier estimate m, s.e. τ	0.81	0.88	0.76	0.73	*0.47*	*0.49*	0.54
Qwen3.5 9B	normal, mean m, sd τ (ours)	1.00	0.81	0.76	0.80	0.26	0.27	0.44
	about m, give or take τ	1.00	0.99	1.00	0.99	0.35	0.43	0.57
	Mean: $m \pm \tau$	1.00	1.08	1.25	1.14	0.19	0.27	0.38
	earlier estimate m, s.e. τ	1.00	0.87	0.86	0.86	0.61	0.75	0.82

Separating stated precision from generating precision. For each task, we take a description value generated either at $c = 36$ or at $c = 1$ and present it with each stated precision in $\{1, 4, 36\}$. The tasks and demonstrations match the full run, and the experiment includes its own no-description prompts at every n (Table 7.5). Repeated conditions reproduce the full run exactly except for gpt-oss-20b. Its distribution readout changes by a median of 0.8 to 1.2 across conditions, with a maximum difference of 39; its written replies change in 12% of cases, with median difference 0. A newer library version is a possible explanation.

The prompt-based reference accepts the stated precision. A value generated at $c = 1$ but stated as $c = 36$ therefore has reference ESS of about -0.8; a model that discounts the stated precision can outperform it in that misspecified condition. For models without reasoning in this control, ESS varies more with the generating precision than with the stated precision. When the value is generated at $c = 1$, each model's ESS remains between -0.1 and 2.0 across all stated precisions. When the value is generated at $c = 36$, each model's ESS is higher, but from stated $c = 1$ to stated $c = 36$ it rises by at most 1.2, whereas reference ESS rises from 2.5 to 36. For example, gpt-oss-20b has ESS 2.7, 2.6, and 2.8 at stated $c = 1$, 4, and 36. Phi-4's ESS even falls, from 12 at stated $c = 1$ to 7.6 at stated $c = 36$, the reverse of the reference order.

Table 7.5: **Sensitivity to stated precision with the value and demonstrations held fixed.** Results use $\sigma = 60$, the full-run tasks, and the readout conventions of Table 7.1. For a value generated at $c = 36$, the table reports its weight at $n = 1$ and summary ESS under each stated c. Results for the 200-task runs of Sections 7.1–7.3 appear in Table 7.6.

	weight, $n = 1$			ESS		
stated c	1	4	36	1	4	36
Reference	0.50	0.80	0.97	2.5	13	36
gpt-oss-20b	0.48	0.47	0.49	2.7	2.6	2.8
Phi-4	0.81	0.75	0.74	12	7.2	7.6
OLMo 2 32B	0.70	0.70	0.70	4.8	4.6	5.3
gpt-oss-20b (written)	0.52	0.46	0.50	2.9	3.5	3.8
Qwen3.5 9B (written)	0.76	0.75	0.80	3.3	3.5	4.0
Gemma 4 12B (written)	0.38	0.29	0.59	1.0	1.0	1.8
Gemma 4 31B (written)	0.43	0.49	0.72	0.2	0.3	1.4

7.10 Reasoning Runs and Execution Settings

Comparisons between modes. The labels "with reasoning" and "without reasoning" refer to generation settings. Statements about individual replies refer to recorded reasoning output. Paired runs use the same model identifier and prompts, but reasoning settings, serving software, readout, retry policy, and access to calculation tools vary across the runs. These comparisons concern the evaluated configurations and do not identify a unique internal mechanism.

Reasoning and benchmark performance. A higher score on Humanity's Last Exam (HLE) does not imply closer agreement with the reference description weights in these configurations. With reasoning, Qwen3.5 9B and gpt-oss-20b have reported HLE scores of 14.9% and 5.3%, respectively, below GPT-6 Sol's 18.4% and Opus 4.6's 19.1% without reasoning in the modes we evaluate (Center for AI Safety et al., 2026; Artificial Analysis, 2026). Nevertheless, their fitted description weights are closer to the Bayesian reference.

Shared evaluation prompts. Most configurations in Figure 7.1 are evaluated on the same 7,800 prompts from the first 200 tasks. These include no-description prompts at $n = 1, 2, 4, \ldots, 64$, with additional points at $n = 40$ and 48 near the $c = 36$ crossing, and every description condition at $n = 1$, 2, and 4. The gpt-oss-20b and Olmo 3 7B Think (Team Olmo, 2025) runs omit the two additional baseline points and contain 7,400 prompts. Results without reasoning for the open-weight models are taken from the full run restricted to these tasks. Paired mode comparisons use the same prompts, although serving environments differ as detailed below. At least 99.7% of replies in each run contain a number.

Open-weight reasoning models. With reasoning, gpt-oss-20b runs in vLLM using greedy decoding, low reasoning effort, and at most 4,096 new tokens. It reaches the final channel in 99.9% of replies and uses 322 tokens on average, and at least 97% of replies reach it in every cell. An earlier run with a limit of 384 tokens finished only 28 to 54% of the replies in the stated-reliability conditions, and the difference between a stated 10% and 50% that this run seemed to show came from which replies finished. Qwen3.5 9B runs through OpenRouter at temperature 0, with reasoning enabled and

a limit of 24,576 new tokens. We pin requests to DeepInfra, which serves the model in bf16, because quantization varies across providers. Of its replies, 2,305 loop to the token limit, a failure its model card warns of. Each is rerun once with the model card's recommended reasoning sampling settings: temperature 1.0, top-p 0.95, top-k 20, and presence penalty 1.5. Eleven remain unfinished and one request fails.

Gemma 4 31B (`gemma-4-31b-it`) runs through the Gemini API at temperature 0. Reasoning is enabled by default, and the model uses 2,613 reasoning tokens on average, with the same 24,576-token limit. Of its replies, 435 reach the limit; all finish after one rerun with the model card's recommended sampling settings: temperature 1.0, top-p 0.95, and top-k 64.

Olmo 3 7B Think (`allenai/Olmo-3-7B-Think`) runs in vLLM with greedy decoding and at most 32,768 new tokens, the limit its model card recommends; at 8,192 tokens, only 12 of 20 pilot replies finished. We rerun the 169 replies that reach this limit once with temperature 0.6 and top-p 0.95; 17 remain unfinished. The model generates 9,215 tokens on average and writes a worked solution before most answers. Its long greedy outputs are sensitive to hardware: an earlier stated-precision run on a different GPU produces the same final number on only half the prompts. We therefore report one run. It averages the demonstrations about as well as the reference but places more weight on the description at $c = 1$ and 4. At $c = 36$, its replies split between the description value alone and other prediction rules, as detailed below.

Models served through command-line agents. GPT-6 Sol (`gpt-6-sol`, the model identifier served by Codex during the runs) runs through the Codex command-line agent in a read-only sandbox, with an empty configuration directory so that no user instructions or settings load, and with every tool that can run code disabled. We evaluate the model with reasoning disabled (no reasoning tokens) and enabled at low effort. An earlier configuration loaded user instructions and allowed shell commands, which the model used in about 8% of replies, mostly to average long demonstration sequences. Without reasoning, that configuration also gave the description much less weight: summary ESS was 1.5 rather than 10.3 at $c = 36$. With reasoning, every summary ESS changes by at most 2.7 between the configurations.

Recorded reasoning output is also more frequent with GPT-6 Sol's earlier configuration at low effort. Without a description, 51%, 63%, and 89% of replies include reasoning output at $n = 1$, 2, and 4, compared with 0.5%, 4%, and 13% in the final

runs. With a stated 50% chance of irrelevance at $n = 1$, every reply includes reasoning output, compared with 74% in the final runs. The wording experiments below used the earlier configuration; the model invoked a shell in 4 of their 5,400 replies.

Claude Opus 4.6 (`claude-opus-4-6`) and Haiku 5.5 (`claude-haiku-5-5`) run through the Claude Code command line, with one prompt per call, no tools, no project or user instructions, and the system prompt "You are a helpful assistant." Opus 4.6 uses high effort in both evaluated modes, with reasoning either enabled or disabled; the disabled mode emits no reasoning tokens. With reasoning enabled, at least 99% of replies in each description condition include reasoning output, with an average of 599 reasoning tokens across all replies. Without a description, only 1%, 24%, and 79% of replies include reasoning output at $n = 1$, 2, and 4, respectively. Haiku 5.5 is evaluated at medium and low effort. Both Claude models use the command line's default sampling settings, which it does not report, and one reply per prompt. Without reasoning, Opus 4.6 still writes visible working in 64 of its 7,800 replies; accounting for these replies changes no summary ESS by more than 0.05.

At medium effort, Haiku 5.5 includes reasoning output in 37 to 88% of replies at $n = 1$ across conditions and in nearly all replies from $n = 2$ onward, with 270 reasoning tokens on average. Its no-description error stays close to the reference through $n = 48$; at $n = 16$, for example, it is 15.8 against 14.8, with two replies far off. Its error rises at $n = 64$, where it divides the sum by 60 instead of 64 in 9 of 200 replies.

Run dates. The full run took place on 2–3 October 2026, and its control conditions and gpt-oss-20b's reasoning run on 20 October. Haiku 5.5 was evaluated on 20 October for the main runs and 4 November for wording, the stated baseline prior, and the semantic task. Opus 4.6, Qwen3.5 9B, Gemma 4 31B, and Olmo 3 7B Think were evaluated on 3–4 November. GPT-6 Sol was evaluated on 3 November for wording and 4 November for the main runs, the stated baseline prior, and the semantic task. The GPT-6 Sol logs do not record a model version beyond the identifier above.

ESS when baseline arithmetic fails. With reasoning, gpt-oss-20b and Qwen3.5 9B nearly match the reference description weight: at $n = 4$ and $c = 36$, their fitted weights are 0.89 and 0.90, compared with 0.90 for the reference. Their errors without descriptions stop falling at larger n. gpt-oss-20b makes errors when averaging long sequences: its no-description root-mean-square error is 19.0 at $n = 32$ and 24.7 at

$n = 64$, compared with reference errors 10.0 and 7.4. At these counts, 37 and 62 replies, respectively, differ from the sample mean by more than 20. Qwen3.5 9B levels off near 11 from $n = 32$ onward (11.2 at $n = 64$), with up to 11 replies per cell that far from the sample mean.

For these models, some description-conditioned errors never meet the baseline curve within the evaluated grid. Table 7.6 therefore also reports, in parentheses, a comparison using the reference's baseline. Specifically, it finds the reference demonstration count t_{ref} that matches the model's description-conditioned error and reports $t_{\text{ref}} - n$. This comparison differs from learner-specific ESS because it uses another predictor's baseline curve. gpt-oss-20b also makes arithmetic errors with descriptions, reducing ESS in those conditions; for example, its summary ESS is 1.3 when the prompt states a 10% chance of irrelevance. About 2% of its replies are more than 40 from the reference prediction, from errors such as exchanging the two variances. On the reference's baseline, its $c = 36$ description is therefore worth 24 demonstrations, well below the reference's 34.6, although its fitted weight is right. A fitted weight hides such errors, whereas ESS does not.

Opus 4.6 and GPT-6 Sol do not exhibit these arithmetic failures. Opus 4.6 has root-mean-square error 7.4 at $n = 64$, matching the reference on these tasks. For these two models, changing between the model and reference baselines changes ESS by at most 1. At low effort, Haiku 5.5 often returns the demonstration without reasoning output. Its description weight at $n = 1$ is consequently low even in conditions whose ESS is close to the reference value.

7.11 Prediction Rules and Relevance Inference

Per-response description weights. At $n = 4$, we compute the per-response description weight $(\hat{y} - \bar{y})/(m - \bar{y})$. This quantity describes one response and differs from the fitted coefficient obtained by regressing predictions across tasks. We use the 61 tasks per cell with $|m - \bar{y}| \geq 30$ and exclude replies without a number. A reply is classified as consistent with a rule when its per-response weight is within 0.05 of that rule's weight. The candidate weights are the Bayesian reference (0.90 for $c = 36$ and 0.20 for $c = 1$); the reference with its two precisions exchanged (0.10 and 0.80); the description counted as one additional observation, $1/(n+1)$; the description alone, with weight 1; and the demonstrations alone, with weight 0. The one-additional-observation

Table 7.6: **Description ESS across reasoning configurations.** Results use the first 200 tasks at $\sigma = 60$. Summary ESS uses each run's own baseline; parentheses report the value computed using the reference baseline where it differs. "10% wrong" and "50% wrong" denote stated probabilities of irrelevance for a description generated at $c = 36$. The last column fixes the generating precision at 36 and states it as 4. Each run names the mode and, in parentheses, the provider's effort setting. An upper bound, $\leq$, indicates that one demonstration alone gives lower error; > 60 denotes a crossing beyond the grid. Reference values use the same tasks and estimator. The open-weight results without reasoning appear as the open points in Figure 7.1 and, on all 2,000 tasks, in Table 7.1. The runs also include three further fixed-value conditions, not shown.

					Stated irrelevance		
Run	$c = 1$	$c = 4$	$c = 36$	Irrelevant	10%	50%	Stated $c = 4$ Generated $c = 36$
Reference	1.0	3.5	34.6	≤ -1.0	27.5	5.7	13.7
GPT-6 Sol, without reasoning	0.9	1.9	10	≤ -1.0	3.3	0.9	2.4
GPT-6 Sol, with reasoning (low)	0.9	3.3	35 (34)	≤ -1.0	30 (29)	4.2	11
Opus 4.6, without reasoning	0.9	2.0	5.8 (5.9)	≤ -1.0	6.1 (6.0)	3.8	3.0
Opus 4.6, with reasoning (high)	1.0	3.5	35	≤ -1.0	29	3.8	13
Gemma 4 31B, with reasoning	1.0	3.5	35	≤ -1.0	21	4.0	14
Qwen3.5 9B, with reasoning	0.9	3.5	> 60 (34)	≤ -1.0	29 (27)	3.9	15 (14)
gpt-oss-20b, with reasoning (low)	0.3	1.0	> 60 (24)	≤ -1.0	1.3 (1.2)	8.8	8.6 (8.4)
Haiku 5.5, with reasoning (medium)	1.1	3.3 (3.5)	28	≤ -1.0	33	3.8 (3.9)	8.5 (8.8)
Haiku 5.5, with reasoning (low)	0.8 (0.7)	3.2 (3.3)	24	≤ -1.0	27 (26)	5.5 (5.6)	8.6 (8.4)
Olmo 3 7B Think, with reasoning	-0.5	0.0	6.2 (6.0)	≤ -1.0	5.9 (5.6)	5.5 (5.3)	15 (13)

rule coincides with the reference at $c = 1$.

With reasoning, Opus 4.6, GPT-6 Sol, Gemma 4 31B, and Haiku 5.5 match the reference rule on 89 to 100% of eligible replies. Without reasoning, 77% of Opus 4.6 replies at $c = 36$ match none of these candidate rules, as do more than half of GPT-6 Sol replies. Olmo 3 7B Think averages demonstrations well but reaches summary ESS only 6.2 at $c = 36$. At $n = 1$, it repeats the description value in 57 to 80% of replies, more often at higher stated precision. At $n = 4$, about half of its eligible replies return the description value alone and a third match the reference. Without a description, its root-mean-square error falls from 56 at $n = 1$ to 10.8 at $n = 32$, compared with 10.0 for the reference. Its shortfall therefore lies in combining the description with demonstrations. Enabling reasoning does not ensure reference behavior in every model.

Within each cell, Haiku 5.5's replies split by whether they include reasoning output. At $n = 1$, replies with reasoning output assign weights 0.50, 0.80, and 0.97 to the relevant $c = 1$, 4, and 36 values, the reference weights to two decimals, at either effort. Replies without reasoning output assign weight 0.00 to 0.05 and repeat the demonstration. Its fitted weight at $n = 1$ mixes the two kinds of reply: 0.44, 0.66, and 0.74 at medium effort. On these three descriptions, 34 to 53% of its replies include reasoning output at low effort, against 72 to 82% at medium effort, so its fitted weight at $n = 1$ is lower at low effort and nearly flat, at 0.36, 0.34, and 0.41. With more demonstrations, more replies include reasoning output, and its weight approaches the reference again: at $n = 4$ and low effort, it is 0.17, 0.48, and 0.86, compared with 0.20, 0.50, and 0.90. Whether a reply includes reasoning output is the model's own choice, so this split describes its replies rather than the result of an experiment.

Averaging with prior weights. For $c = 36$, we compare the Bayesian reference with a predictor that conditions each component on the demonstrations and averages the resulting posterior means with prior weights p and $1 - p$. In the scalar approximation, this gives weight $p\,36/(36 + n)$ to m. Table 7.7 reports fitted description weights under the original warning, which leaves the baseline prior unspecified. At the stated 50%, GPT-6 Sol, Gemma 4 31B, and Qwen3.5 9B with reasoning are close to the predictions obtained with prior weights. Opus 4.6 and Haiku 5.5 at medium effort fall below the Bayesian reference weight by $n = 4$. Haiku 5.5 at low effort comes closest to the reference; its replies often contain no reasoning output at $n = 1$. gpt-oss-20b assigns nearly the weight of a relevant description whatever the stated probability.

Table 7.7: **Description weights under stated unreliability.** Fitted coefficients on the $c = 36$ description value for the first 200 tasks, with stated irrelevance probability $1 - p$. All model rows use reasoning. "Prior weights" uses weight $p\,36/(36 + n)$; the reference updates the relevance probability using a uniform baseline prior over 100 to 999. Bootstrap standard errors are at most 0.03 at $n = 4$ in the stated-50% condition and at most 0.06 overall, except for gpt-oss-20b, whose maximum is 0.09. At the stated 50%, several reasoning configurations closely track the predictions obtained by averaging with prior weights.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Prior weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol	0.49	0.46	0.40	0.90	0.86	0.71
Opus 4.6	0.56	0.43	0.40	0.93	0.86	0.77
Gemma 4 31B	0.49	0.46	0.45	0.89	0.68	0.76
Qwen3.5 9B	0.49	0.46	0.43	0.88	0.84	0.80
Haiku 5.5, medium	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, low	0.49	0.62	0.58	0.42	0.94	0.88
gpt-oss-20b	0.89	0.87	0.82	0.95	0.87	0.90

Sensitivity to the unspecified baseline prior. The original prompt states only that, if the description is wrong, "the mean is unknown". A Bayesian relevance update compares the marginal likelihoods of the demonstrations under the two component priors. The prompt therefore does not determine a unique posterior relevance probability.

For the $c = 36$ description with a stated 50% chance of irrelevance, the reference's fitted weights on the first 200 tasks at $n = 1$, 2, 4, 8, and 16 are 0.58, 0.61, 0.64, 0.58, and 0.54 with the uniform baseline prior over 100 to 999. They are 0.49, 0.49, 0.53, 0.49, and 0.45 with the actual task prior $\mathcal{N}(500, 100^2)$; 0.52, 0.54, 0.58, 0.53, and 0.49 with $\mathcal{N}(500, 200^2)$; and 0.72, 0.74, 0.74, 0.67, and 0.60 with a uniform baseline prior over 0 to 2,000. Averaging with prior weights gives 0.49, 0.47, 0.45, 0.41, and 0.35.

At four demonstrations, the reference's fitted description weight exceeds that obtained with prior weights for every baseline prior examined. The model predictions and the comparison using prior weights are unchanged; only the Bayesian reference varies with the assumed baseline prior. Under the actual task prior, updating the relevance probability changes the fitted weight little before four demonstrations.

Stating the baseline prior. To remove this ambiguity, we rerun the stated-reliability condition on the same 200 tasks, at $c = 36$, stated irrelevance probabilities of 10% and 50%, and $n = 1$, 2, and 4. The prompt now states the reference's uniform baseline prior over 100 to 999. We use GPT-6 Sol at low effort and Haiku 5.5 at medium effort, with 1,200 prompts each. Table 7.8 compares their fitted description weights with those from the original runs.

At the stated 50%, the reference and prior-weighted predictions differ by at least 5 on 123 to 155 tasks per cell. On these tasks, the share of replies closer to the reference than to the prior-weighted prediction rises from 0–5% to 92–97% for GPT-6 Sol, and from 21–41% to 87–95% for Haiku 5.5. At the stated 10%, at most 7 tasks per cell separate the two rules by this amount, so the comparison is uninformative there.

A constant increase in the description weight could also make replies closer to the reference, because its posterior relevance probability exceeds the stated 50% on most tasks. To test whether the weights vary with the evidence, we use each reply's *normalized description weight*: its per-response weight divided by $c/(c+n)$, restricting to tasks with $|m - \bar{y}| \geq 30$. For the scalar reference, this equals its posterior relevance probability π_n, up to the negligible boundary effects discussed above. For model

predictions, the ratio can lie outside $[0, 1]$ and does not identify an internal probability. Here π_n ranges from 0.08 to 0.88 as the demonstrations provide different evidence for relevance; averaging with prior weights keeps the normalized weight constant.

We regress the models' normalized weights on π_n, pooling $n = 1$, 2, and 4 at the stated 50% (269 replies per model). With the baseline prior stated, the slopes are 0.96 [0.84, 1.04] for GPT-6 Sol and 0.98 [0.82, 1.15] for Haiku 5.5, compared with 0.01 [−0.09, 0.11] and 0.22 [−0.01, 0.45] under the original warning. The 95% intervals resample the 153 tasks. The corresponding intercepts with the prior stated are 0.01 and 0.03, and R^2 is 0.57 and 0.33.

The agreement also holds when the demonstrations count against the description. Among the 23 replies with π_n below 0.5 (mean 0.33), the mean normalized weights are 0.30 and 0.38 for the two models with the baseline prior stated, compared with 0.48 and 0.50 under the original warning. Both models therefore vary their description weights across tasks in approximately the amount predicted by Bayesian model averaging.

Summary ESS uses each model's no-description curve from the main run. At the stated 50%, averaging with prior weights gives 4.4 and the Bayesian reference gives 5.7. Stating the baseline prior raises ESS from 4.2 to 5.4 for GPT-6 Sol and from 3.8 to 4.2 for Haiku 5.5. At the stated 10%, it changes from 29.7 to 25.0 for GPT-6 Sol and from 32.6 to 15.1 for Haiku 5.5, compared with 27.5 for the reference. The two models' fitted weights agree within 0.01 in this condition (Table 7.8). Haiku's lower ESS therefore does not reflect lower average reliance on the description; it comes from a minority of replies far from the reference, some of which repeat the demonstration.

7.12 Wording with Reasoning Enabled

GPT-6 Sol. We evaluate the three alternative wordings at low effort for the three relevant descriptions and $n = 1$, 2, and 4, using the first 200 tasks. Of 5,400 replies, 5,399 contain a number. With "An earlier estimate put its mean at m, with a standard error of τ", every reply includes reasoning output, and the fitted description weight is within 0.01 of the reference in every cell.

Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", reasoning output is absent from 24 to 51% and 11 to 50% of replies per cell, respectively. Of those replies, 83% and 87% repeat the description value. At $c = 1$ and $n = 1$, the fitted description weights are consequently 0.89 and 1.00, compared with reference weight 0.50. Among

Table 7.8: **Stating the baseline prior changes description weights.** Fitted coefficients on the $c = 36$ description value for the first 200 tasks, as in Table 7.7. "Unstated" denotes the original warning, "the mean is unknown"; "stated" specifies that the mean is otherwise uniform over 100 to 999, as the reference assumes. GPT-6 Sol uses low effort and Haiku 5.5 uses medium effort.

	50% wrong			10% wrong		
	$n = 1$	2	4	1	2	4
Reference	0.58	0.61	0.64	0.88	0.87	0.84
Prior weights	0.49	0.47	0.45	0.88	0.85	0.81
GPT-6 Sol, unstated	0.49	0.46	0.40	0.90	0.86	0.71
GPT-6 Sol, stated	0.58	0.59	0.59	0.88	0.83	0.80
Haiku 5.5, unstated	0.55	0.56	0.46	0.79	0.93	0.89
Haiku 5.5, stated	0.62	0.60	0.63	0.89	0.84	0.80

replies with reasoning output, 68% and 43% are within 1 of the reference prediction, while 36% and 68% repeat the description value. These categories overlap at $c = 36$, where the description value and reference prediction can be close.

Haiku 5.5. We evaluate the same alternatives at medium effort; all 5,400 replies contain a number. With "An earlier estimate put its mean at m, with a standard error of τ", at least 71% of replies in each cell include reasoning output; 99% of these lie within 1 of the reference prediction. Under "Its mean is about m, give or take τ" and "Mean: $m \pm \tau$", reasoning output is absent from 4 to 36% and 2 to 74% of replies per cell, respectively. Even among replies with reasoning output, 71% and 91% repeat the description value, while 31% and 20% lie within 1 of the reference prediction. At $c = 1$ and $n = 4$, its description weights are 0.72 and 0.93, compared with reference weight 0.20.

All wordings provide a value m and a scale τ, but "give or take" and "$\pm$" do not identify τ as a prior standard deviation or specify a distribution for the mean. The comparison varies the explicitness of the probabilistic specification as well as its wording. The results show that accurate combination of descriptions and demonstrations under explicit statistical statements need not extend to these less explicit formulations.

7.13 Details of the Semantic Task

Task. A message has six binary properties: its sender is outside the organization; it requests a payment; it concerns travel; it requests computer access; it has a deadline within a day; and it contains personal contact information. Each message states every property as present or absent, using one of three phrasings for each value, such as "I'm a contractor from another company." or "I'm on the sales team here." Properties are sampled independently with probability one half, phrasings are sampled uniformly, and sentences appear in random order.

A policy uses exactly two properties and requires each to be present or absent. A message goes into folder A when both requirements hold, and into B otherwise. There are $\binom{6}{2}2^2 = 60$ such policies. The prompt describes this family and states that the company's policy was chosen uniformly from it. It then provides a policy note, n previously filed messages with their folders, and four new messages. The model is asked for each new message's probability of going into A, as a number between 0 and 1, on four lines from "1: <probability>" to "4: <probability>". Without reasoning, GPT-6 Sol identifies each property correctly in 97–100% of 96 messages; all three misreadings concern one phrasing, "I'm not traveling for this."

Policy notes. A broad note reads, for example, "Policy note: whether it concerns travel is one of the two properties the policy uses," leaving 20 possible policies. A specific note reads "Policy note: to go into A, a message must request a payment. The other property the policy uses is whether it concerns travel," leaving 2. These notes communicate constraints in ordinary language; the model must determine which policies satisfy them.

The unreliable note adds "This note has a 50% chance of being wrong, in which case it says nothing about the policy, which is then any of the 60, equally likely." In the generator, the note is relevant with probability one half and otherwise describes an independently drawn policy. The latter, irrelevant branch occurs in 92 of the 200 tasks. An independently generated note can agree with the actual policy by chance. Thus, *relevance* denotes the latent generating branch, whereas *compatibility* denotes whether a policy satisfies the note; the two are not interchangeable.

Reference and ESS. For a note asserted to be relevant, the reference predicts the fraction of policies consistent with the note and the filed messages that assign the new message to A. For an unreliable note, it averages this prediction with the prediction from policies consistent with the filed messages alone, weighting the two by the posterior relevance probability. The prior relevance probability is 0.5; the posterior probability is obtained using the likelihood of the demonstrations under each branch. On 3,000 simulated tasks, the reference's predictive probabilities are calibrated in every condition, and its posterior probability that the unreliable note is relevant matches the frequency of relevance: where it exceeds 0.7, it averages 0.80, and the note is relevant in 80% of cases.

The loss is the squared error between the predicted probability of A and the true folder indicator, averaged over the four query messages. ESS follows Definition 3.1, using each learner's no-note curve at $n = 0$, 2, 4, 6, 8, 12, 16, 20, 24, and 32, with linear interpolation in n. On simulated tasks, this grid changes the reference ESS by at most 0.6 relative to evaluation at every integer demonstration count.

Runs. We use 200 tasks, each defined by a sampled policy, and 19 conditions: no note at the ten demonstration counts above, and each note at $n = 0$, 2, and 4. This gives 3,800 prompts per configuration. Haiku 5.5 runs at medium effort; GPT-6 Sol runs without reasoning and at low effort. The command-line agents follow Section 7.10, with no tools, and no reply calls a tool. Fourteen replies from GPT-6 Sol at low effort contain no probability and are excluded; one Haiku 5.5 reply omits one of its four probabilities. Every Haiku 5.5 reply includes reasoning output, with a median of 3,008 reasoning tokens. For GPT-6 Sol at low effort, 91% of replies include reasoning output, with a median of 142 reasoning tokens.

Results. Table 7.9 reports each note's ESS. With reasoning, both models are within 2 demonstrations of the reference in every condition except GPT-6 Sol's specific note at $n = 4$: its ESS is 26.1, compared with 14.8 for the reference. Its loss with this note, 0.008, is close to the reference's 0.007, but its no-note curve flattens beyond $n = 16$, with loss 0.022 at $n = 20$, compared with 0.004 for the reference. It consequently needs more demonstrations on its own baseline curve to match the note's error reduction.

For the unreliable note, we compare predictions with the Bayesian reference and with a comparator that averages the two component predictions using their prior

weights. This comparator retains the stated 50% weight unless the filed messages rule out every policy compatible with the note; in that case, it assigns zero weight to the note-conditioned component. We use query messages on which the two predictions differ by at least 0.05: about 150 at $n = 2$ and 200 at $n = 4$, out of 800. The share of predictions closer to the reference is 90% at $n = 2$ and 99% at $n = 4$ for Haiku 5.5; 77% and 87% for GPT-6 Sol at low effort; and 45% at both counts for GPT-6 Sol without reasoning.

Without reasoning, GPT-6 Sol learns slowly from demonstrations. Its loss without a note is still 0.078 at $n = 32$, compared with 0.000 for the reference at the reported precision. Measured against this weaker baseline, a note can have an ESS as large as the reference's or larger. For example, the specific note has ESS 13.5 at $n = 2$, although the model's loss with that note is 0.130, compared with 0.027 for the reference. This illustrates why learner-specific ESS must be interpreted alongside absolute prediction error.

Table 7.9: **ESS of policy notes in the semantic task.** Results use 200 tasks and each learner's own curve without a note. "≤ 0" means that the note did not lower loss; "> 28" means that the no-note curve does not reach the note-conditioned loss by $n = 32$. On 6,000 simulated tasks, the reference ESS is 9.6 and 13.2 for the specific note at $n = 2$ and 4, and 2.5 and 2.9 for the unreliable note. Models are compared with the reference on the same 200 tasks. Paired 95% bootstrap intervals over tasks (1,000 replicates) for the specific note at $n = 2$ are: reference [9.7, 12.9], Haiku 5.5 [8.7, 11.5], GPT-6 Sol at low effort [7.8, 12.7], and GPT-6 Sol without reasoning [3.7, 21.9].

Note	n	Reference	Haiku 5.5 medium	GPT-6 Sol low	GPT-6 Sol no reasoning
Broad	0	0.0	0.0	≤ 0	≤ 0
	2	1.4	1.3	1.3	0.7
	4	1.8	1.6	1.7	8.6
Specific	0	3.7	3.6	3.3	3.7
	2	11.4	9.9	9.5	13.5
	4	14.8	14.5	26.1	> 28
Unreliable	0	1.3	1.3	1.6	1.7
	2	3.0	2.6	3.2	1.7
	4	5.0	4.9	5.1	2.7

Chapter 8

Discussion

A task description's value depends on its precision, its reliability, and the demonstrations already available. In our Gaussian model, a reliable description can replace many demonstrations by constraining the task across directions. Uncertain relevance imposes a finite ESS ceiling before demonstrations are observed. Demonstrations can then supply evidence about relevance or constrain directions left unspecified by a partial description, increasing its ESS.

Meta-trained Transformers approach the Bayes-optimal benchmark, while their response to reliability shifts reflects training. With reasoning enabled, most tested pretrained language models produce description weights closer to the Bayesian reference. For the two models tested with an explicit irrelevant-component prior, agreement with posterior weighting extends from numerical predictions to semantic policy constraints. Incomplete or informal probabilistic specifications produce different behavior, making the stated model assumptions consequential to the comparison.

The analysis offers a controlled interpretation of why descriptions sometimes substitute for many examples (Le Scao & Rush, 2021; Honda et al., 2025) and sometimes have limited influence (Webson & Pavlick, 2022; Gupta et al., 2025; Gloaguen et al., 2026). It identifies distinct possible causes: low specificity, uncertain relevance, and a learner's limited use of the available information. Establishing which cause explains a practical prompting result requires evaluating that setting directly.

8.1 Scope and Limitations

The theory assumes Gaussian linear regression, squared error, and a particular mixture model of irrelevance. Learner-specific ESS measures improvement relative to a learner's own baseline and should be read alongside absolute error. The language-model tasks are small and controlled; the main numerical comparison uses 200 tasks, and the explicit-prior and semantic experiments cover two models. Some reasoning comparisons also differ in serving, decoding, or readout settings. GPT-6 Sol's results without reasoning depend on its agent configuration. Agreement with a Bayesian reference concerns predictions and does not identify the model's internal computation. Natural descriptions with unstated uncertainty, more structured tasks, and richer or heavy-tailed priors (Dalal & Hall, 1983; O'Hagan & Pericchi, 2012) would test how far these findings extend.

The wording test with reasoning enabled also covers only two models. The original reliability prompts leave the baseline prior unspecified, so the Bayesian reference for them depends on an assumed prior; we report several choices (Chapter 7). The networks are trained only at $d = 5$ and, for the higher-dimension experiment, at $d = 10$. Because the numerical language-model task is scalar, it has no random-design dimension term and no partial description, and the language models we test are post-trained chat models. Models without reasoning are evaluated through either the distribution readout or the written readout, so comparisons across these models mix differences between models with differences between readouts.

8.2 Future Work

We outline four directions for future work. The first two follow from the limitations and findings above, the third from the descriptions studied here, and the fourth from the networks of Chapter 6. None of them is established in this thesis.

Partial descriptions and the dimension term in language models. A precise partial description can become more valuable as demonstrations resolve what it leaves unspecified (Proposition 4.2), whereas the ESS of a reliable complete description falls toward $c - a_0^{-1}$ (Corollary 5.2). At $d = 16$, $a_0 = 10$, and $c = 100$, ESS increases from 8.1 without demonstrations to 36.2 with 100 demonstrations for $k = 8$, and falls from

117 to 109 for $k = 16$ (Table 4.1). The networks of Chapter 6 capture this increase with $k = 3$ of $d = 5$ directions described, and their ESS at $c = 10$ is larger at $d = 10$ than at $d = 5$. The numerical language-model task, however, has a single unknown mean. A task with several unknown quantities, of which the description states only some, would test whether a language model's ESS has a dimension term, and whether a partial description becomes more valuable to it as demonstrations accumulate.

Updating reliability. With reasoning, GPT-6 Sol and Haiku 5.5 update relevance from the demonstrations once the prompt states the baseline prior: at a stated 50% chance of irrelevance, regressing their normalized description weights on the reference's posterior relevance probability gives slopes of 0.96 and 0.98. At a stated 10% chance, however, Haiku 5.5's ESS falls to 15.1, against 27.5 for the reference and 25.0 for GPT-6 Sol. The two models' fitted weights agree within 0.01 in this condition, so Haiku 5.5's shortfall comes from a minority of replies far from the reference, some of which repeat the demonstration. This rerun covers two models, $c = 36$, and $n \leq 4$, and its descriptions remain relevant. Three questions remain open. Do the other reasoning models update relevance once the baseline prior is stated? Why do some of Haiku 5.5's replies depart so far from the reference at a small stated chance of irrelevance? And do reasoning models also update relevance under actual mixtures of relevant and irrelevant descriptions, which Table 7.3 evaluates only without reasoning?

Structural descriptions. Apart from the semantic task, the descriptions studied here state where the task lies, through a center m with a stated precision. Many instructions instead state what kind of task it is, for example that only three features matter, or that the answer is monotone in an input. Such a description narrows the hypothesis class rather than pointing at one task. One case of this kind is already covered. A description that confines w to a known s-dimensional subspace states the other $k = d - s$ directions exactly, so it is a partial description of those directions in the limit $c \to \infty$. A heuristic calculation suggests that such a description keeps its value as demonstrations accumulate. Under a flat baseline prior, it would lower the Bayes-optimal error from about d/n to about s/n, so its ESS would be about $n(d/s - 1)$, which grows with the number of demonstrations. Structures that do not fix any direction, such as a sparse task whose relevant features are unknown, or a monotone one, are not covered. The definition of ESS and the network and

language-model tests apply to them as well, and the semantic task of Section 7.4 gives a first test of one in a language model: its broad note names a property the policy uses without stating its required value, and with reasoning, Haiku 5.5 and GPT-6 Sol value it at 1.3 demonstrations at $n = 2$, against 1.4 for the reference (Table 7.9). Moreover, a wrong structural description would leave the Bayesian predictor assuming relevance with a bias that no number of demonstrations removes.

Where a network stores its trust. The meta-trained networks inherit their trust from training (Chapter 6). A natural next step is to locate the components of a network that hold this trust, and to test by intervention whether changing them changes the network's implied trust without retraining. Such a study would connect the behavioral measure of this thesis to the mechanisms by which networks learn in context.

Bibliography

Abdin, M., Aneja, J., Behl, H., et al. Phi-4 technical report. Technical report, Microsoft, 2024. arXiv:2412.08905.

Akata, E., Voudouris, K., Fortuin, V., and Schulz, E. In-context function learning in large language models. In *AISTATS*, 2026.

Akyürek, E., Schuurmans, D., Andreas, J., Ma, T., and Zhou, D. What learning algorithm is in-context learning? Investigations with linear models. In *ICLR*, 2023.

Anthropic. System card: Claude Haiku 5.5. https://www.anthropic.com/claude-haiku-5-5-system-card, 2026a.

Anthropic. System card: Claude Opus 4.6. https://anthropic.com/claude-opus-4-6-system-card, 2026b.

Arora, A., Jurafsky, D., Potts, C., and Goodman, N. D. Bayesian scaling laws for in-context learning. In *COLM*, 2025.

Artificial Analysis. Humanity's Last Exam evaluation. https://artificialanalysis.ai/evaluations/humanitys-last-exam, 2026. Accessed 4 November 2026.

Brown, T. B. et al. Language models are few-shot learners. In *NeurIPS*, 2020.

Center for AI Safety, Scale AI, and HLE Contributors Consortium. A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649:1139–1146, 2026. doi: 10.1038/s41586-025-09962-4.

Cook, J., Sapora, S., Ahmadian, A., Khan, A., Rocktäschel, T., Foerster, J., and Ruis, L. Programming by backprop: an instruction is worth 100 examples when finetuning LLMs. In *ICLR*, 2026.

Cover, T. M. and Thomas, J. A. *Elements of Information Theory*. Wiley, second edition, 2006.

Dalal, S. R. and Hall, W. J. Approximating priors by mixtures of natural conjugate priors. *Journal of the Royal Statistical Society: Series B (Methodological)*, 45(2): 278–286, 1983.

Evans, M. and Moshonov, H. Checking for prior-data conflict. *Bayesian Analysis*, 1 (4):893–914, 2006.

Falck, F., Wang, Z., and Holmes, C. Is in-context learning in large language models Bayesian? A martingale perspective. In *ICML*, 2024.

Filbry, L., Melo, L. C., and Gal, Y. From prompts to posteriors: Can LLM agents perform Bayesian inference in context? *arXiv:2610.29331*, 2026.

Garg, S., Tsipras, D., Liang, P., and Valiant, G. What can Transformers learn in-context? A case study of simple function classes. In *NeurIPS*, 2022.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., and Rubin, D. B. *Bayesian Data Analysis*. CRC Press, third edition, 2013.

Gemma Team. Gemma 4 technical report. *arXiv:2607.02770*, 2026.

Genewein, T., Wenliang, L. K., Grau-Moya, J., Ruoss, A., Orseau, L., and Hutter, M. Understanding prompt tuning and in-context learning via meta-learning. In *NeurIPS*, 2025.

Gloaguen, T., Mündler-Sasahara, N., Müller, M. N., Raychev, V., and Vechev, M. Evaluating AGENTS.md: are repository-level context files helpful for coding agents? *arXiv:2602.11988*, 2026.

Gupta, R., Corona, R., Ge, J., Wang, E., Klein, D., Darrell, T., and Chan, D. M. Enough coin flips can make LLMs act Bayesian. In *ACL*, 2025.

He, Z., Qiu, T., Shirado, H., and Sap, M. Martingale Score: An unsupervised metric for Bayesian rationality in LLM reasoning. In *NeurIPS*, 2025.

Hoeting, J. A., Madigan, D., Raftery, A. E., and Volinsky, C. T. Bayesian model averaging: A tutorial. *Statistical Science*, 14(4):382–417, 1999. doi: 10.1214/ss/1009212519.

Honda, U., Murakami, S., and Zhang, P. Distilling many-shot in-context learning into a cheat sheet. In *Findings of EMNLP*, 2025.

Huang, R. and Ge, R. Task descriptors help Transformers learn linear models in-context. In *ICLR*, 2025.

Kwa, T., West, B., Becker, J., Deng, A., Garcia, K., Hasin, M., Jawhar, S., Kinniment, M., Rush, N., Von Arx, S., et al. Measuring AI ability to complete long software tasks. *arXiv:2503.14499*, 2025.

Le Scao, T. and Rush, A. M. How many data points is a prompt worth? In *NAACL*, 2021.

Lin, Z. and Lee, K. Dual operating modes of in-context learning. In *ICML*, 2024.

Lin, Z., Bharti, S. K., and Lee, K. In-context learning with hypothesis-class guidance. *arXiv:2502.19787*, 2025.

Liu, T. J. B., Boullé, N., Sarfati, R., and Earls, C. J. Density estimation with LLMs: a geometric investigation of in-context learning trajectories. In *ICLR*, 2025.

Mao, Y., He, J., and Chen, C. From prompts to templates: A systematic prompt template analysis for real-world LLMapps. In *FSE Industry Track*, 2025. arXiv:2504.02052.

Min, S., Lyu, X., Holtzman, A., Artetxe, M., Lewis, M., Hajishirzi, H., and Zettlemoyer, L. Rethinking the role of demonstrations: What makes in-context learning work? In *EMNLP*, 2022.

MiniCPM Team. MiniCPM5-2B. https://huggingface.co/openbmb/MiniCPM5-2B, 2026. Model card; the series is described in *MiniCPM4: Ultra-Efficient LLMs on End Devices*, arXiv:2506.07900.

Morita, S., Thall, P. F., and Müller, P. Determining the effective sample size of a parametric prior. *Biometrics*, 64(2):595–602, 2008.

Muirhead, R. J. *Aspects of Multivariate Statistical Theory*. Wiley, 1982.

Müller, S., Hollmann, N., Pineda Arango, S., Grabocka, J., and Hutter, F. Transformers can do Bayesian inference. In *ICLR*, 2022.

Neuenschwander, B., Weber, S., Schmidli, H., and O'Hagan, A. Predictively consistent prior effective sample sizes. *Biometrics*, 76(2):578–587, 2020.

O'Hagan, A. and Pericchi, L. Bayesian heavy-tailed models and conflict resolution: A review. *Brazilian Journal of Probability and Statistics*, 26(4):372–401, 2012.

OpenAI. gpt-oss-120b & gpt-oss-20b model card. Technical report, OpenAI, 2025. arXiv:2508.10925.

OpenAI. GPT-6 Sol and GPT-6 Luna: October 2026 update. https://deployment safety.openai.com/gpt-6-october, 2026.

Panwar, M., Ahuja, K., and Goyal, N. In-context learning through the Bayesian prism. In *ICLR*, 2024.

Patsakis, C., Argyropoulos, V., and Alepis, E. Configuration, not conscience: A large-scale empirical study of LLM system prompts. In *ICTAI*, 2026. arXiv:2609.31575.

Pichon, F., Dubois, D., and Denœux, T. Relevance and truthfulness in information correction and fusion. *International Journal of Approximate Reasoning*, 53(2): 159–175, 2012.

Qiu, L., Sha, F., Allen, K., Kim, Y., Linzen, T., and van Steenkiste, S. Bayesian teaching enables probabilistic reasoning in large language models. *Nature Communications*, 17:1238, 2026.

Qwen Team. Qwen3.5-9B. https://huggingface.co/Qwen/Qwen3.5-9B, 2026. Model card.

Raventós, A., Paul, M., Chen, F., and Ganguli, S. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *NeurIPS*, 2023.

Reimherr, M., Meng, X.-L., and Nicolae, D. L. Prior sample size extensions for assessing prior impact and prior-likelihood discordance. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83(3):413–437, 2021.

Requeima, J., Bronskill, J., Choi, D., Turner, R. E., and Duvenaud, D. LLM processes: numerical predictive distributions conditioned on natural language. In *NeurIPS*, 2024.

Reznik, Y. A. The optimal discounting parameter of the power prior under predictive log-loss. *arXiv:2608.12159*, 2026.

Schmidli, H., Gsteiger, S., Roychoudhury, S., O'Hagan, A., Spiegelhalter, D., and Neuenschwander, B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics*, 70(4):1023–1032, 2014.

Selby, D., Iwashita, Y., Spriestersbach, K., Saad, M., Bappert, D., Warrier, A., Mukherjee, S., Kise, K., and Vollmer, S. Had enough of experts? Quantitative knowledge retrieval from large language models. *Stat*, 14:e70054, 2025.

Team Olmo. Olmo 3. *arXiv preprint arXiv:2512.13961*, 2025.

Team OLMo, Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., et al. 2 OLMo 2 Furious. In *COLM*, 2025. arXiv:2501.00656.

Tong, X., Zeng, Y., and Zhang, J. Demonstrations, CoT, and prompting: a theoretical analysis of ICL. *arXiv:2603.19611*, 2026.

von Oswald, J., Niklasson, E., Randazzo, E., Sacramento, J., Mordvintsev, A., Zhmoginov, A., and Vladymyrov, M. Transformers learn in-context by gradient descent. In *ICML*, 2023.

von Rosen, D. Moments for the inverted Wishart distribution. *Scandinavian Journal of Statistics*, 15(2):97–109, 1988.

Webson, A. and Pavlick, E. Do prompt-based models really understand the meaning of their prompts? In *NAACL*, 2022.

Wei, J., Wei, J., Tay, Y., Tran, D., Webson, A., Lu, Y., Chen, X., Liu, H., Huang, D., Zhou, D., and Ma, T. Larger language models do in-context learning differently. *arXiv:2303.03846*, 2023.

Weller, O., Lourie, N., Gardner, M., and Peters, M. E. Learning from task descriptions. In *EMNLP*, pp. 1361–1375, 2020. doi: 10.18653/v1/2020.emnlp-main.105.

Wenliang, L. K., Ruoss, A., Grau-Moya, J., Hutter, M., and Genewein, T. Why is prompting hard? Understanding prompts on binary sequence predictors. In *AISTATS*, 2026. PMLR 300:2134–2142.

Wiesenfarth, M. and Calderazzo, S. Quantification of prior impact in terms of effective current sample size. *Biometrics*, 76(1):326–336, 2020.

Xie, S. M., Raghunathan, A., Liang, P., and Ma, T. An explanation of in-context learning as implicit Bayesian inference. In *ICLR*, 2022.

Zhu, Q., Oermann, E. K., and Cho, K. Multi-task Bayesian in-context learning. In *ICML*, 2026. arXiv:2606.20538.

Appendix A

Language-Model Results at Every Number of Demonstrations

This appendix collects the complete language-model tables of Chapter 7. Tables A.1–A.3 report ESS and description weights at every evaluated demonstration count for the full run without reasoning: Tables A.1 and A.2 report ESS, and Table A.3 reports description weights for the models with a distribution readout. Tables A.1 and A.3 each continue onto a second page.

Table A.1: **ESS by description condition and demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is the numeric response rate averaged over n. "10% wrong" and "50% wrong" denote stated irrelevance probabilities. The censoring marker $\leq$ marks error above the first baseline point; $> 64 - n$ means that the description-conditioned error lies below the entire baseline curve through 64 demonstrations.

Predictor	Description	number	$n = 1$	$n = 2$	$n = 4$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
Reference	$\sigma = 60$, $c = 1$, relevant		1.2	1.3	1.3	1.3	1.2	1.5	> 0
	$\sigma = 60$, $c = 1$, irrelevant		≤ 0.0	-0.5	-0.8	-1.2	-1.7	-1.5	-1.7
	$\sigma = 60$, $c = 4$, relevant		4.5	4.4	3.9	4.6	4.6	5.2	> 0
	$\sigma = 60$, $c = 4$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	-6.2	-12	-20	-32
	$\sigma = 60$, $c = 36$, relevant		36	36	37	38	39	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant		≤ 0.0	≤ -1.0	≤ -3.0	≤ -7.0	≤ -15	≤ -31	-62
	$\sigma = 30$, $c = 1$, relevant		1.0	1.2	1.4	1.6	1.4	1.5	> 0
	$\sigma = 30$, $c = 4$, relevant		4.5	4.7	5.0	5.1	4.9	5.6	> 0
	$\sigma = 30$, $c = 36$, relevant		42	43	43	43	42	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong		3.8	3.9	3.6	4.4	4.5	5.1	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong		≤ 0.0	-0.7	-1.2	-1.6	-2.2	-1.9	-2.2
	$\sigma = 60$, $c = 36$, relevant, 10% wrong		13	22	26	35	38	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong		≤ 0.0	-0.9	-1.7	-2.8	-5.4	-6.9	-10
	$\sigma = 60$, $c = 4$, relevant, 50% wrong		1.9	2.2	2.9	3.7	4.1	4.5	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong		≤ 0.0	-0.2	-0.4	-0.5	-0.8	-0.6	-0.7
	$\sigma = 60$, $c = 36$, relevant, 50% wrong		3.1	5.8	12	21	31	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong		≤ 0.0	-0.3	-0.7	-1.2	-2.4	-2.8	-4.5
gpt-oss-20b	$\sigma = 60$, $c = 1$, relevant	0.98	1.0	0.9	-0.1	-1.5	-9.2	-25	-57
	$\sigma = 60$, $c = 1$, irrelevant	0.98	≤ 0.0	-0.3	-1.5	-3.6	-10	-25	-59
	$\sigma = 60$, $c = 4$, relevant	0.98	2.3	1.7	2.0	2.9	-7.5	-24	-48
	$\sigma = 60$, $c = 4$, irrelevant	0.97	≤ 0.0	-0.3	-1.4	-3.7	-11	-26	-59
	$\sigma = 60$, $c = 36$, relevant	0.98	2.8	2.3	3.8	17	3.7	-12	-17
	$\sigma = 60$, $c = 36$, irrelevant	0.98	≤ 0.0	-0.5	-1.5	-3.7	-11	-26	-60
	$\sigma = 30$, $c = 1$, relevant	0.99	0.8	0.4	-0.3	-1.8	-9.9	-26	-59
	$\sigma = 30$, $c = 4$, relevant	0.98	1.2	1.3	1.9	3.4	-8.1	-24	-44
	$\sigma = 30$, $c = 36$, relevant	0.99	1.6	2.3	4.7	31	3.0	-13	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	2.2	1.8	0.0	-0.2	-8.1	-24	-52
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.6	-1.1	-2.2	-9.6	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.97	2.4	2.4	0.5	2.8	-3.6	-20	-33
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	≤ 0.0	-0.7	-1.1	-2.3	-9.7	-25	-59
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.95	2.3	1.7	0.1	-0.5	-8.3	-24	-53
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.94	≤ 0.0	-0.5	-1.1	-2.3	-9.7	-25	-58
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.95	2.8	2.3	0.7	1.2	-4.5	-21	-37
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.95	≤ 0.0	-0.6	-1.1	-2.3	-9.7	-25	-59
Phi-4	$\sigma = 60$, $c = 1$, relevant	0.87	0.8	0.7	-0.2	-0.4	-1.0	-3.1	-35
	$\sigma = 60$, $c = 1$, irrelevant	0.86	≤ 0.0	≤ -1.0	≤ -3.0	-4.9	-6.1	-8.8	-40
	$\sigma = 60$, $c = 4$, relevant	0.85	5.8	4.1	2.4	2.1	0.7	-4.6	-33
	$\sigma = 60$, $c = 4$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.7	-6.2	-13	-42
	$\sigma = 60$, $c = 36$, relevant	0.86	15	7.6	4.7	5.6	8.5	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	0.84	≤ 0.0	≤ -1.0	≤ -3.0	-4.5	-6.4	-12	-42
	$\sigma = 30$, $c = 1$, relevant	0.87	1.1	0.8	-0.3	-0.3	-1.0	-1.8	-15
	$\sigma = 30$, $c = 4$, relevant	0.86	4.8	3.5	2.0	1.6	0.6	-1.9	-2.2
	$\sigma = 30$, $c = 36$, relevant	0.88	9.7	6.0	3.8	4.4	8.4	17	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.87	6.0	5.5	5.3	8.3	15	> 32	> 0

	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.85 \leq$	$0.0 \leq$	-1.0	-2.1	-1.2	2.9	-3.2	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.87	10	8.7	8.2	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.86 \leq$	$0.0 \leq$	-1.0	-2.4	-1.7	1.3	-4.7	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.87	6.0	6.4	7.7	17	> 48	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.85 \leq$	$0.0 \leq$	-1.0	-1.3	1.5	16	> 32	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.88	9.2	9.7	11	> 56	> 48	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.86 \leq$	$0.0 \leq$	-1.0	-1.8	0.6	13	> 32	> 0
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	1.9	1.8	2.0	2.3	5.8	> 32	> 0
	$\sigma = 60$, $c = 1$, irrelevant	$0.89 \leq$	0.0	-0.5	-0.2	-0.2	1.7	25	> 0
	$\sigma = 60$, $c = 4$, relevant	0.91	5.8	3.4	3.4	4.5	9.3	> 32	> 0
	$\sigma = 60$, $c = 4$, irrelevant	$0.90 \leq$	0.0	-0.7	-0.3	-0.5	0.3	17	> 0
	$\sigma = 60$, $c = 36$, relevant	0.91	11	4.6	5.3	6.8	15	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant	$0.90 \leq$	$0.0 \leq$	-1.0	-0.6	-0.5	-0.2	15	> 0
	$\sigma = 30$, $c = 1$, relevant	0.92	1.8	1.9	1.7	1.7	4.4	24	> 0
	$\sigma = 30$, $c = 4$, relevant	0.92	4.8	3.3	3.1	4.0	9.5	> 32	> 0
	$\sigma = 30$, $c = 36$, relevant	0.93	7.0	4.4	4.0	5.9	14	> 32	> 0
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	6.0	4.6	3.8	4.5	5.8	23	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	$0.96 \leq$	0.0	-0.9	0.0	0.0	-0.4	4.1	> 0
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	12	5.9	6.0	6.3	8.9	31	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	$0.96 \leq$	$0.0 \leq$	-1.0	-0.3	-0.1	-0.8	0.3	> 0
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	5.3	4.2	3.5	4.0	5.4	27	> 0
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	$0.92 \leq$	0.0	-0.5	0.1	0.1	0.0	11	> 0
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	7.8	5.4	4.9	5.6	7.9	> 32	> 0
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	$0.92 \leq$	0.0	-0.8	-0.1	0.0	-0.3	7.8	> 0

Table A.2: **ESS by description condition and demonstration count: written readout.** Results cover the three written-readout models in the full run without reasoning. Replies without a qualifying number are excluded. Columns and reference values follow Table A.1.

Predictor	Description	number	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Qwen3.5 9B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.2	0.3	−0.9	−2.8	−9.3	−24	−51
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.7	−4.8	−9.7	−23	−49
	$\sigma=60$, $c=4$, relevant	1.00	3.9	2.7	1.3	−0.7	−6.2	−18	−44
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	≤ -1.0	−1.7	−3.8	−9.4	−24	−49
	$\sigma=60$, $c=36$, relevant	1.00	6.5	4.0	2.8	6.9	17	>32	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−2.8	−5.1	−10	−24	−51
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.2	−0.7	−2.5	−8.7	−22	−49
	$\sigma=30$, $c=4$, relevant	1.00	3.4	2.7	1.5	−0.3	−4.5	−14	−40
	$\sigma=30$, $c=36$, relevant	1.00	4.7	4.3	3.0	9.2	14	>32	>0
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	3.8	−0.4	−0.7	−1.8	−8.5	−21	−47
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−4.9	−10	−24	−50
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	5.3	0.4	0.6	−0.3	−3.3	−4.2	−6.9
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−5.2	−10	−25	−53
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	4.5	−0.1	−0.4	−1.9	−9.0	−23	−50
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−6.2	−12	−26	−54
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	7.3	0.8	1.0	−0.1	−4.0	−4.2	−12
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	≤ -1.0	≤ -3.0	−7.0	−13	−27	−56
Gemma 4 12B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.0	0.0	−0.1	−0.6	−3.2	−5.4	−21
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.4	−1.3	−1.3	−4.4	−6.8	−24
	$\sigma=60$, $c=4$, relevant	1.00	0.3	0.5	1.4	1.1	−0.6	−2.7	−13
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.4	−1.1	−1.2	−4.1	−6.9	−23
	$\sigma=60$, $c=36$, relevant	1.00	1.8	1.2	2.7	3.6	3.6	4.9	>0
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	−0.8	−1.3	−1.3	−4.9	−7.9	−26
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	−0.2	−0.1	−0.5	−3.2	−7.9	−27
	$\sigma=30$, $c=4$, relevant	1.00	0.7	0.8	2.1	1.5	−0.2	−4.6	−18
	$\sigma=30$, $c=36$, relevant	1.00	4.0	1.7	3.4	3.7	2.4	−1.8	−8.9
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.7	1.4	0.4	−1.0	−3.1	−7.3
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.4	−0.5	−2.2	−5.4	−13
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	0.6	1.3	2.3	1.6	0.7	−0.6	−1.3
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	−0.2	−0.5	−0.5	−2.5	−5.8	−15
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	1.0	0.6	1.7	1.0	−0.1	−1.7	−4.5
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.4	−0.4	−1.3	−4.1	−9.6
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	5.1	1.0	2.7	2.3	2.4	1.1	>0
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	−0.3	−0.6	−0.4	−1.9	−4.8	−13
Gemma 4 31B (written)	$\sigma=60$, $c=1$, relevant	1.00	0.1	0.0	−0.1	−0.5	−1.4	−5.1	−13
	$\sigma=60$, $c=1$, irrelevant	1.00	≤ 0.0	−0.1	−0.1	−0.5	−1.0	−4.5	−11
	$\sigma=60$, $c=4$, relevant	1.00	0.8	0.1	0.2	−0.5	−1.0	−4.3	−12
	$\sigma=60$, $c=4$, irrelevant	1.00	≤ 0.0	−0.2	−0.2	−0.4	−1.0	−4.4	−10
	$\sigma=60$, $c=36$, relevant	1.00	2.6	0.6	1.4	0.9	2.9	3.4	−2.9
	$\sigma=60$, $c=36$, irrelevant	1.00	≤ 0.0	≤ -1.0	−0.8	−0.6	−1.1	−2.7	−4.8
	$\sigma=30$, $c=1$, relevant	1.00	≤ 0.0	0.0	−0.1	−0.7	−1.2	−3.0	−8.1
	$\sigma=30$, $c=4$, relevant	1.00	0.5	0.0	0.1	−0.5	−1.3	−3.4	−8.3
	$\sigma=30$, $c=36$, relevant	1.00	4.3	1.5	2.1	2.3	3.3	3.3	−0.4
	$\sigma=60$, $c=4$, relevant, 10% wrong	1.00	0.1	0.0	0.0	−0.7	−1.9	−5.0	−15
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.2	−4.1	−9.5
	$\sigma=60$, $c=36$, relevant, 10% wrong	1.00	1.0	0.0	0.6	−0.3	−1.4	−4.8	−12
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−0.9	−3.0	−6.2
	$\sigma=60$, $c=4$, relevant, 50% wrong	1.00	0.2	0.0	0.0	−0.7	−1.7	−4.6	−17
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.2	−0.4	−1.2	−3.6	−12
	$\sigma=60$, $c=36$, relevant, 50% wrong	1.00	1.0	0.0	0.5	−0.3	−1.5	−4.5	−15
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	1.00	≤ 0.0	0.0	−0.1	−0.4	−1.0	−2.9	−8.5

Table A.3: **Description weights by demonstration count: distribution readout.** Results cover the prompt-based reference and the three distribution-readout models in the full run without reasoning. "Number" is defined as in Table A.1; cells are italicized when the numeric response rate is below three quarters at the corresponding n.

Predictor	Description	number	$n=0$	$n=1$	$n=2$	$n=4$	$n=8$	$n=16$	$n=32$	$n=64$
Reference	$\sigma=60$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=1$, irrelevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=60$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=4$, irrelevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=60$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=36$, irrelevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=30$, $c=1$, relevant		1.00	0.50	0.33	0.20	0.11	0.06	0.03	0.02
	$\sigma=30$, $c=4$, relevant		1.00	0.80	0.67	0.50	0.33	0.20	0.11	0.06
	$\sigma=30$, $c=36$, relevant		1.00	0.97	0.95	0.90	0.82	0.69	0.53	0.36
	$\sigma=60$, $c=4$, relevant, 10% wrong		0.90	0.71	0.59	0.45	0.30	0.19	0.10	0.05
	$\sigma=60$, $c=4$, irrelevant, 10% wrong		0.90	0.27	0.14	0.08	0.04	0.02	0.01	0.00
	$\sigma=60$, $c=36$, relevant, 10% wrong		0.90	0.84	0.84	0.82	0.77	0.65	0.51	0.34
	$\sigma=60$, $c=36$, irrelevant, 10% wrong		0.90	0.26	0.14	0.07	0.03	0.01	0.00	0.00
	$\sigma=60$, $c=4$, relevant, 50% wrong		0.50	0.46	0.39	0.32	0.22	0.14	0.08	0.04
	$\sigma=60$, $c=4$, irrelevant, 50% wrong		0.50	0.12	0.07	0.03	0.02	0.01	0.00	0.00
	$\sigma=60$, $c=36$, relevant, 50% wrong		0.50	0.55	0.58	0.60	0.59	0.53	0.42	0.29
	$\sigma=60$, $c=36$, irrelevant, 50% wrong		0.50	0.12	0.06	0.03	0.01	0.00	0.00	0.00
gpt-oss-20b	$\sigma=60$, $c=1$, relevant	0.98	0.94	0.53	0.29	0.23	0.16	0.12	0.12	0.23
	$\sigma=60$, $c=1$, irrelevant	0.98	0.93	0.59	0.25	0.18	0.09	0.05	0.04	0.10
	$\sigma=60$, $c=4$, relevant	0.98	0.92	0.48	0.32	0.26	0.21	0.19	0.18	0.31
	$\sigma=60$, $c=4$, irrelevant	0.97	0.92	0.57	0.25	0.16	0.08	0.04	0.04	0.09
	$\sigma=60$, $c=36$, relevant	0.98	0.94	0.49	0.35	0.30	0.25	0.23	0.22	0.35
	$\sigma=60$, $c=36$, irrelevant	0.98	0.95	0.60	0.25	0.15	0.07	0.04	0.04	0.09
	$\sigma=30$, $c=1$, relevant	0.99	0.98	0.43	0.23	0.17	0.12	0.09	0.11	0.21
	$\sigma=30$, $c=4$, relevant	0.98	0.98	0.38	0.28	0.23	0.19	0.18	0.19	0.31
	$\sigma=30$, $c=36$, relevant	0.99	0.99	0.37	0.32	0.27	0.23	0.22	0.24	0.37
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.96	0.91	0.45	0.33	0.15	0.13	0.12	0.13	0.22
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.96	0.91	0.55	0.34	0.11	0.05	0.02	0.02	0.04
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.97	0.92	0.44	0.34	0.17	0.15	0.15	0.16	0.25
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.96	0.92	0.53	0.34	0.10	0.04	0.02	0.02	0.04
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.95	0.79	0.46	0.32	0.16	0.12	0.12	0.12	0.20
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.94	0.79	0.53	0.32	0.11	0.04	0.02	0.01	0.04
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.95	0.83	0.46	0.32	0.17	0.14	0.15	0.15	0.23
	$\sigma=60$, $c=36$, irrelevant, 50% wrong	0.95	0.83	0.53	0.31	0.11	0.04	0.02	0.02	0.04
Phi-4	$\sigma=60$, $c=1$, relevant	0.87	*0.97*	0.82	0.62	0.43	0.28	0.18	0.13	0.12
	$\sigma=60$, $c=1$, irrelevant	0.86	*0.97*	0.83	0.64	0.42	0.22	0.11	0.06	0.05
	$\sigma=60$, $c=4$, relevant	0.85	*0.98*	0.76	0.55	0.37	0.26	0.19	0.15	0.16
	$\sigma=60$, $c=4$, irrelevant	0.84	*0.98*	0.79	0.56	0.36	0.18	0.09	0.06	0.05
	$\sigma=60$, $c=36$, relevant	0.86	*0.99*	0.74	0.53	0.38	0.29	0.23	0.20	0.20
	$\sigma=60$, $c=36$, irrelevant	0.84	*0.99*	0.78	0.53	0.33	0.16	0.09	0.06	0.04
	$\sigma=30$, $c=1$, relevant	0.87	*0.99*	0.77	0.58	0.38	0.22	0.13	0.09	0.09
	$\sigma=30$, $c=4$, relevant	0.86	*1.00*	0.69	0.51	0.34	0.23	0.16	0.13	0.13
	$\sigma=30$, $c=36$, relevant	0.88	*1.00*	0.65	0.48	0.32	0.23	0.18	0.17	0.16
	$\sigma=60$, $c=4$, relevant, 10% wrong	0.87	*0.94*	0.62	0.46	0.30	0.21	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 10% wrong	0.85	*0.95*	0.62	0.44	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 10% wrong	0.87	*0.95*	0.61	0.46	0.33	0.25	0.20	0.18	0.16
	$\sigma=60$, $c=36$, irrelevant, 10% wrong	0.86	*0.96*	0.61	0.43	0.26	0.13	0.06	0.04	0.03
	$\sigma=60$, $c=4$, relevant, 50% wrong	0.87	*0.92*	0.57	0.43	0.29	0.20	0.15	0.13	0.11
	$\sigma=60$, $c=4$, irrelevant, 50% wrong	0.85	*0.92*	0.57	0.42	0.24	0.12	0.06	0.04	0.03
	$\sigma=60$, $c=36$, relevant, 50% wrong	0.88	*0.93*	0.56	0.44	0.33	0.24	0.21	0.19	0.17

	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.86	*0.94*	0.57	0.41	0.24	0.12	0.06	0.04	0.03
OLMo 2 32B	$\sigma = 60$, $c = 1$, relevant	0.90	*0.99*	0.72	0.33	0.25	0.16	0.09	0.05	0.05
	$\sigma = 60$, $c = 1$, irrelevant	0.89	*0.99*	0.77	0.29	0.16	0.09	0.05	0.02	0.02
	$\sigma = 60$, $c = 4$, relevant	0.91	*1.00*	0.70	0.37	0.30	0.21	0.14	0.08	0.07
	$\sigma = 60$, $c = 4$, irrelevant	0.90	*1.00*	0.75	0.29	0.15	0.08	0.04	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant	0.91	*1.00*	0.70	0.40	0.34	0.26	0.19	0.12	0.10
	$\sigma = 60$, $c = 36$, irrelevant	0.90	*1.00*	0.76	0.29	0.17	0.08	0.04	0.02	0.02
	$\sigma = 30$, $c = 1$, relevant	0.92	*0.99*	0.69	0.32	0.23	0.14	0.09	0.04	0.04
	$\sigma = 30$, $c = 4$, relevant	0.92	*1.00*	0.65	0.34	0.27	0.19	0.13	0.06	0.06
	$\sigma = 30$, $c = 36$, relevant	0.93	*1.00*	0.64	0.36	0.32	0.23	0.16	0.08	0.09
	$\sigma = 60$, $c = 4$, relevant, 10% wrong	0.96	1.01	0.72	0.42	0.29	0.18	0.11	0.09	0.05
	$\sigma = 60$, $c = 4$, irrelevant, 10% wrong	0.96	1.01	0.81	0.32	0.14	0.06	0.03	0.02	0.01
	$\sigma = 60$, $c = 36$, relevant, 10% wrong	0.96	1.01	0.71	0.44	0.33	0.23	0.15	0.13	0.07
	$\sigma = 60$, $c = 36$, irrelevant, 10% wrong	0.96	1.01	0.82	0.33	0.15	0.07	0.03	0.02	0.01
	$\sigma = 60$, $c = 4$, relevant, 50% wrong	0.93	1.02	0.65	0.39	0.27	0.16	0.09	0.07	0.04
	$\sigma = 60$, $c = 4$, irrelevant, 50% wrong	0.92	1.01	0.71	0.29	0.13	0.06	0.02	0.01	0.01
	$\sigma = 60$, $c = 36$, relevant, 50% wrong	0.93	1.02	0.64	0.42	0.31	0.21	0.13	0.11	0.06
	$\sigma = 60$, $c = 36$, irrelevant, 50% wrong	0.92	1.02	0.72	0.30	0.14	0.06	0.03	0.02	0.01